# Dorey's Resource Book

## Differentiation

Derivatives of exponential, logarithmic and trigonometric functions

## Contents

Episode 1. The derivative of $e^{f(x)}$

Episode 2. Derivatives of $a^x$, products and quotients

Episode 3. Tangents and normals with exponentials

Episode 4. The derivative of $\ln x$ and $\ln f(x)$

Episode 5. $\log_a x$, combined rules and tangents

Episode 6. Derivatives of $\sin x$, $\cos x$ and $\tan x$

Episode 7. Trig functions with combined rules

Episode 8. Derivatives of the reciprocal trig functions

Episode 9. Mixed differentiation: choosing the rule

# 1 The derivative of $e^{f(x)}$

ANCHOR QUESTION

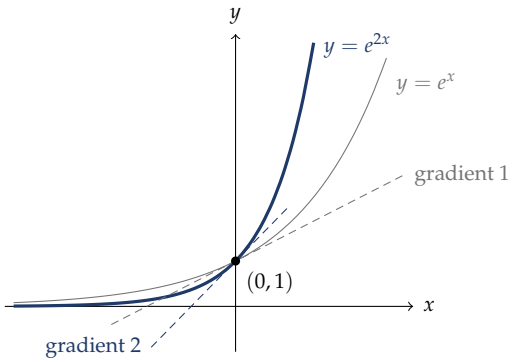
*Differentiate* $y = e^{x^2-5x}$.

SYLLABUS MAV-12-04

- Apply the rules of differentiation to find the derivative of $f(x) = ke^{ax}$, where $k$ and $a$ are constants
- Use the chain rule to prove $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where $a$ and $b$ are constants and $a \neq 0$
- Use the chain rule to differentiate functions of the form $e^{f(x)}$

NOTES

- $\frac{d}{dx}\left(e^x\right) = e^x$.
- $\frac{d}{dx}\left(e^{f(x)}\right) = f'(x)\,e^{f(x)}$. This is on the reference sheet.
- The power stays exactly the same. Multiply by the derivative of the power.
- Special case: $\frac{d}{dx}\left(e^{ax+b}\right) = a\,e^{ax+b}$, and $\frac{d}{dx}\left(ke^{ax}\right) = ka\,e^{ax}$.
- Proof: let $u = ax + b$, so $y = e^u$. Then $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.
- $x$ in the power of $e$: use this rule. $x$ in the base, as in $x^e$: use the power rule. No $x$, as in $e^4$: a constant, derivative 0.
- Rewrite first with index laws: $\frac{1}{e^{kx}} = e^{-kx}$, $e^a e^b = e^{a+b}$, $\sqrt{e^x} = e^{x/2}$.
- $e^{f(x)} > 0$ for every $x$, so $f'(x)\,e^{f(x)} = 0$ only when $f'(x) = 0$.




WORKED EXAMPLE

*Differentiate* $y = e^{3x^2-x}$.

$f(x) = 3x^2 - x$, so $f'(x) = 6x - 1$. the power and its derivative

$$\frac{dy}{dx} = f'(x)\,e^{f(x)}$$
$$= (6x - 1)\,e^{3x^2-x}$$

## Tier 1 · the derivative of $e^{f(x)}$

---

**1.** For each function, write the power $f(x)$, then find $f'(x)$.

a) $e^{9x}$

b) $e^{2-x}$

c) $e^{x^4-3x}$

**2.** Differentiate:

a) $y = e^{7x}$

b) $y = e^{x^2+1}$

c) $y = e^{x^3-6x}$

**3.** Differentiate:

a) $y = e^{1-5x}$

b) $y = e^{\sqrt{x}}$

c) $y = e^4$

## Tier 2 · a coefficient in front

---

**4.** Differentiate:

a) $y = 4e^{3x}$

b) $y = -2e^{5x+1}$

c) $y = \frac{1}{2}e^{6x}$

**5.** Differentiate:

a) $y = 3e^{x^2}$

b) $y = -e^{2x^2-x}$

**6.** Show that $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = 2e^{8x-2}$.

## Tier 3 · sums and differences

---

**7.** Differentiate:

a) $y = e^{4x} + e^{-4x}$

b) $y = 3e^{2x} - x^3$

**8.** Differentiate $y = 2e^{-3x} - 5e^{x} + 7x$.

**9.** Differentiate $y = x^2 - 4x + e^{2-x}$.

## Tier 4 · rewrite with index laws first

**10.** Differentiate:

a) $y = \dfrac{1}{e^x}$

b) $y = \dfrac{4}{e^{2x}}$

**11.** Differentiate:

a) $y = e^x \cdot e^{3x}$

b) $y = \left(e^{2x}\right)^3$

**12.** Differentiate:

a) $y = \sqrt{e^{4x}}$

b) $y = (e^x + e^{-x})^2$, by expanding first

## Tier 5 · exact values of the derivative

**13.** If $f(x) = e^{2x+1}$, find the exact value of $f'(0)$.

**14.** If $f(x) = e^{x^2-4}$, find the exact value of $f'(2)$.

**15.** Find the gradient of the tangent to $y = 3e^{-x} + e^x$ at the point where $x = \ln 3$.

## Tier 6 · the chain rule on the outside

**16.** Differentiate $y = (e^x + 1)^3$.

**17.** Differentiate $y = \left(2 - e^{3x}\right)^4$.

**18.** Differentiate $y = \sqrt{1 + e^{2x}}$.

## Tier 7 · show that, then hence

**19.** Let $y = e^{x^2} \cdot e^{-2x}$.

a) Show that $\dfrac{dy}{dx} = 2(x-1)\,e^{x^2-2x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

**20.** Let $y = \dfrac{e^{2x^2}}{e^{8x}}$.

a) Show that $\dfrac{dy}{dx} = 4(x-2)\,e^{2x^2-8x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

**21.** Let $f(x) = e^{x^3-3x}$.

a) Find $f'(x)$ in fully factorised form.

b) Find the exact coordinates of the points where the tangent is horizontal.

## Conceptual

**22.** A student writes $\frac{d}{dx}\left(e^{3x}\right) = 3x\,e^{3x-1}$. Explain the error and give the correct derivative.

**23.** Use the chain rule to prove that $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where $a$ and $b$ are constants.

**24.** Explain why the graph of $y = 5e^{2x}$ is increasing for every value of $x$.

**25.** The curve $y = e^{x^2+kx}$ has a horizontal tangent at $x = 3$. Find the value of $k$.

**26.** Let $y = 5e^{2x}$.

(a) Find $\frac{dy}{dx}$. (1 mark)

(b) Show that $\frac{dy}{dx} = 2y$. (1 mark)

(c) Hence find the gradient of the curve at the point where $y = 10$. (1 mark)

## Revision

**27.** Differentiate $y = (3x-2)^4$.

**28.** Differentiate $y = \sqrt{x} + \frac{2}{x}$.

**29.** Use the product rule to differentiate $y = x^2(x+3)^3$, giving your answer in factorised form.

**30.** Use the quotient rule to differentiate $y = \frac{2x+1}{x-3}$.

**31.** Find the equation of the tangent to $y = x^3 - 2x$ at the point where $x = 1$.

**32.** Find the gradient of the normal to $y = x^2 + 4x$ at the point where $x = -1$.

**33.** Simplify $\frac{e^{2x} \times e^{-5x}}{e^x}$.

**34.** Solve $e^{2x} = 9$, giving the exact answer.

**35.** Evaluate $\log_2 32$.

**36.** Simplify $\ln\left(e^3\right) + \ln 1$.

**37.** Solve $\log_3(x+1) = 2$.

**38.** Write $\ln\left(x^2(x+1)\right)$ as a sum of logarithms.

**39.** Find the exact value of $\sin\frac{2\pi}{3}$.

**40.** Solve $\sin x = \frac{1}{2}$ for $0 \leq x \leq 2\pi$.

**41.** Convert $\dfrac{5\pi}{6}$ radians to degrees.

**42.** Simplify $1 - \cos^2\theta$.

**43.** State the domain of $f(x) = \dfrac{1}{\sqrt{x+2}}$.

**44.** If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$.

**45.** Show that $f(x) = x^3 - x$ is an odd function.

**46.** Solve $x^2 - 6x + 5 < 0$.

**47.** Find the inverse function of $f(x) = 3x - 4$.

**48.** Describe the transformations that map $y = f(x)$ onto $y = 2f(x-1)$.

**49.** Events $A$ and $B$ are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cap B)$.

**50.** Two fair dice are rolled. Find the probability that the sum is 7.

**51.** Solve $|x + 3| = 2x$.

**52.** Find the angle of inclination of the tangent to $y = x^2$ at the point where $x = \frac{1}{2}$.

# 2 Derivatives of $a^x$, products and quotients

ANCHOR QUESTION

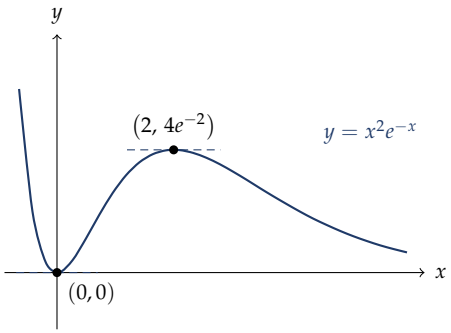
*Differentiate $y = x^2e^{3x}$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Prove and use the formula $\frac{d}{dx}(a^x) = (\ln a)\,a^x$, where $a$ is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course

NOTES

- Any positive number is a power of $e$: $a = e^{\ln a}$, so $a^x = e^{x\ln a}$.
- $\frac{d}{dx}(a^x) = (\ln a)\,a^x$. Proof: $\frac{d}{dx}\left(e^{x\ln a}\right) = \ln a \cdot e^{x\ln a} = (\ln a)\,a^x$.
- $\ln a$ is just a number. When $a = e$, $\ln e = 1$ and the rule gives back $\frac{d}{dx}(e^x) = e^x$.
- Product rule: if $y = uv$, then $\frac{dy}{dx} = u'v + uv'$.
- Quotient rule: if $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$. Both rules are on the reference sheet.
- After the product rule, both terms share the exponential: take it out as a common factor, e.g. $2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$.
- An exponential is never 0, so a factorised derivative is 0 only where its other factors are 0.




WORKED EXAMPLE

*Differentiate $y = x^2e^{5x}$, giving your answer in factorised form.*

$u = x^2,\ u' = 2x; \quad v = e^{5x},\ v' = 5e^{5x}$ — name the parts

$\frac{dy}{dx} = u'v + uv' = 2xe^{5x} + 5x^2e^{5x}$ — product rule

$= xe^{5x}(2 + 5x)$ — common factor $xe^{5x}$

## Tier 1 · the product rule with $e$

**53.** Write as a power of $e$.

a) $9^x$

b) $2^{5x}$

**54.** Factorise fully.

a) $3x^2e^{2x} + 2x^3e^{2x}$

b) $e^{3x} - 3xe^{3x}$

**55.** Differentiate:

a) $y = 6^x$

b) $y = 5 \cdot 2^x$

c) $y = 3^{4x}$

**56.** Differentiate, giving each answer in factorised form.

a) $y = xe^x$

b) $y = x^2e^{4x}$

**57.** Differentiate $y = x^4e^x$, giving your answer in factorised form.

## Tier 2 · a negative power

**58.** Differentiate $y = (3x - 1)e^{-x}$, giving your answer in factorised form.

**59.** Differentiate $y = xe^{-2x}$, giving your answer in factorised form.

**60.** Differentiate $y = x^2e^{-3x}$, giving your answer in factorised form.

## Tier 3 · the quotient rule

**61.** Differentiate $y = \dfrac{e^{3x}}{x}$.

**62.** Differentiate $y = \dfrac{x}{e^{2x}}$, simplifying your answer.

**63.** Differentiate $y = \dfrac{e^x}{x+1}$.

## Tier 4 · products with $a^x$

**64.** Differentiate $y = x \cdot 2^x$.

**65.** Differentiate $y = x^2 \cdot 5^x$, giving your answer in factorised form.

**66.** Differentiate $y = 2^x e^x$.

## Tier 5 · quotients of exponentials

**67.** Differentiate $y = \dfrac{e^x}{e^x + 2}$.

**68.** Differentiate $y = \dfrac{e^x - 1}{e^x + 1}$.

**69.** Differentiate $y = \dfrac{e^{2x}}{1 + e^{2x}}$.

## Tier 6 · horizontal tangents

**70.** Find the exact coordinates of the point on $y = xe^{-x}$ where the tangent is horizontal.

**71.** Find the exact coordinates of the points on $y = x^2 e^x$ where the tangent is horizontal.

**72.** Find the exact coordinates of the point on $y = \dfrac{e^x}{x}$ where the tangent is horizontal.

## Conceptual

**73.** Prove that $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$, where $a > 0$.

**74.** A student writes $\dfrac{d}{dx}\left(2^x\right) = x \cdot 2^{x-1}$. Explain the error and give the correct derivative.

**75.** Show that $\dfrac{d}{dx}\left(xe^x\right) = (x+1)e^x$, and hence find the gradient of $y = xe^x$ at the origin.

**76.** Find the exact gradient of the tangent to $y = xe^{2x}$ at the point where $x = 1$.

**77.** Let $f(x) = xe^{-x}$.

(a) Show that $f'(x) = (1 - x)e^{-x}$. (1 mark)

(b) Explain why $f(x)$ is increasing for $x < 1$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

## Revision

**78.** Differentiate $y = e^{4x-1}$.

**79.** Differentiate $y = (e^x + 2)^3$.

**80.** If $f(x) = e^{x^2-1}$, find $f'(1)$.

**81.** Differentiate $y = \dfrac{1}{e^{2x}}$.

**82.** Evaluate $\ln\left(e^{-2}\right)$.

**83.** Solve $3^{x+1} = 27$.

**84.** Solve $e^x = 5$, giving the exact answer.

**85.** Evaluate $\log_2 8 + \log_2 4$.

**86.** Write $\ln\left(x^3 y\right)$ as a sum of logarithms.

**87.** Differentiate $y = (2x - 5)^3$.

**88.** Differentiate $y = \sqrt{3x + 1}$.

**89.** Find the equation of the tangent to $y = x^2 - 4x$ at the point where $x = 3$.

**90.** Find the gradient of the normal to $y = x^3$ at the point where $x = 1$.

**91.** Solve $x^2 - x - 6 = 0$.

**92.** Solve $2x^2 + 5x - 3 = 0$.

**93.** Find the exact value of $\cos\dfrac{\pi}{3}$.

**94.** Solve $\tan x = 1$ for $0 \le x \le 2\pi$.

**95.** Convert $\dfrac{2\pi}{5}$ radians to degrees.

**96.** State the domain of $y = \ln(x - 3)$.

**97.** If $f(x) = x^2$ and $g(x) = x + 3$, find $g(f(2))$.

**98.** Is $f(x) = x^4 + x^2$ even, odd or neither? Give a reason.

**99.** Solve $|2x + 1| < 5$.

**100.** Events $A$ and $B$ are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.5$. Find $P(A \cup B)$.

**101.** Find the midpoint of $(-2,\ 5)$ and $(4,\ -1)$.

**102.** Simplify $\dfrac{x^2 - 9}{x + 3}$.

# 3 Tangents and normals with exponentials

ANCHOR QUESTION

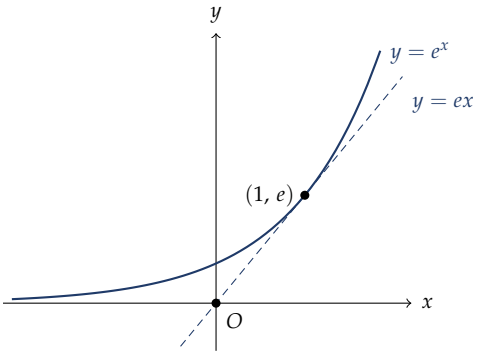
*Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 1$.*

SYLLABUS MAV-12-04

- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- The gradient of the tangent at $x = a$ is $f'(a)$.
- The normal is perpendicular to the tangent, so its gradient is $-\dfrac{1}{f'(a)}$.
- Find the point with the original function: $y = f(a)$. Find the gradient with the derivative.
- Use $y - y_1 = m(x - x_1)$, then rearrange to $y = mx + c$ or general form.
- Leave $e$ in exact answers: $e^0 = 1$, $e^1 = e$, $e^{\ln k} = k$.




WORKED EXAMPLE

*Find the equation of the tangent to $y = e^{4x}$ at the point where $x = 1$.*

$x = 1$: $y = e^4$, so the point is $(1, e^4)$. — point from $f$

$\dfrac{dy}{dx} = 4e^{4x}$, so $m = 4e^4$. — gradient from $f'$

$y - e^4 = 4e^4(x - 1)$ — point–gradient form

$y = 4e^4x - 3e^4$

## Tier 1 · tangents to $y = e^{kx}$

**103.** Find the exact value.

a) $e^{4x}$ at $x = 1$

b) $3e^{1-x}$ at $x = 1$

c) $xe^{3x}$ at $x = 1$

**104.** The tangent has gradient $m$. Find the gradient of the normal.

a) $m = 5e$

b) $m = e^4$

c) $m = -\dfrac{3}{e}$

**105.** Find the equation of the tangent to $y = e^x$ at the point where $x = 2$.

**106.** Find the equation of the tangent to $y = e^{5x}$ at the point where $x = 0$.

**107.** Find the equation of the tangent to $y = 2e^{3x}$ at the point where $x = 1$.

## Tier 2 · normals

**108.** Find the equation of the normal to $y = e^{3x}$ at the point where $x = 0$. Give your answer in general form.

**109.** Find the equation of the normal to $y = e^x$ at the point where $x = 1$. Give your answer in general form.

**110.** Find the equation of the normal to $y = e^{-2x}$ at the point where $x = 0$. Give your answer in general form.

## Tier 3 · tangents that need the product rule

**111.** Find the equation of the tangent to $y = xe^{2x}$ at the origin.

**112.** Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$.

**113.** Find the equation of the tangent to $y = x^2e^x$ at the point where $x = 1$.

### Tier 4 · normals in general form

**114.** Find the equation of the normal to $y = e^{x^2}$ at the point where $x = -1$. Give your answer in general form.

**115.** Find the equation of the normal to $y = e^{2x-2}$ at the point where $x = 1$. Give your answer in general form.

**116.** Find the equation of the normal to $y = e^{x^2-4}$ at the point where $x = 2$. Give your answer in general form.

### Tier 5 · show that

**117.** Show that the tangent to $y = e^{2x}$ at the point where $x = a$ meets the $x$-axis at $x = a - \frac{1}{2}$.

**118.** Show that the tangent to $y = e^{-x}$ at the point where $x = a$ meets the $x$-axis at $x = a + 1$.

**119.** Show that the tangent to $y = e^x$ at the point where $x = a$ meets the $y$-axis at $y = (1 - a)e^a$.

### Tier 6 · finding the point

**120.** The tangent to $y = e^{2x}$ at the point where $x = a$ passes through the origin. Find $a$ and the equation of the tangent.

**121.** Find the exact coordinates of the point on $y = e^x$ where the tangent has gradient 3.

**122.** Find the exact coordinates of the point on $y = e^{3x}$ where the normal has gradient $-\frac{1}{6}$.

### Conceptual

**123.** Explain why no tangent to $y = e^x$ is horizontal.

**124.** Show that the tangents to $y = e^x$ and $y = e^{-x}$ at the point $(0, 1)$ are perpendicular.

**125.** Find where the normal to $y = e^x$ at the point where $x = 0$ crosses the $x$-axis.

**126.** A student finds the tangent to $y = e^{2x}$ at $x = 1$ using the point $(1, 2e^2)$. Explain the error.

**127.** Let $f(x) = e^{2x}$.

(a) Find the equation of the tangent to $y = f(x)$ at the point where $x = 0$. (2 marks)

(b) Find where this tangent meets the $x$-axis. (1 mark)

## Revision

**128.** Differentiate $y = xe^{3x}$, giving your answer in factorised form.

**129.** Differentiate $y = 4^x$.

**130.** Differentiate $y = \dfrac{e^x}{x}$.

**131.** Differentiate $y = e^{x^3}$.

**132.** Differentiate $y = (1 - e^x)^2$.

**133.** Solve $e^{2x} = e^5$.

**134.** Solve $2^x = 10$, giving the exact answer.

**135.** Simplify $e^{\ln 7}$.

**136.** Evaluate $\log_3 81$.

**137.** Find the gradient of the line $2x - 5y + 1 = 0$.

**138.** Find the equation of the line through $(-1, 3)$ perpendicular to $y = 2x + 1$. Give your answer in general form.

**139.** Find the distance between $(1, 2)$ and $(4, 6)$.

**140.** Differentiate $y = \left(x^2 + 1\right)^4$.

**141.** Differentiate $y = \dfrac{x}{x - 1}$.

**142.** Solve $x^2 = 4x$.

**143.** Solve $x^2 + 2x - 2 = 0$, giving exact answers.

**144.** Find the exact value of $\sin \dfrac{3\pi}{4}$.

**145.** Solve $2 \cos x = 1$ for $0 \leq x \leq 2\pi$.

**146.** State the range of $y = e^x + 2$.

**147.** State the equation of the horizontal asymptote of $y = 3 - e^{-x}$.

**148.** Find the inverse function of $f(x) = e^x$.

**149.** Solve $|x - 3| = 5$.

**150.** If $P(A) = 0.6$, find $P\left(\overline{A}\right)$.

**151.** Expand $(x + 2)^3$.

**152.** Simplify $\dfrac{\left(2x^3\right)^2}{4x^2}$.

# 4 The derivative of $\ln x$ and $\ln f(x)$

ANCHOR QUESTION

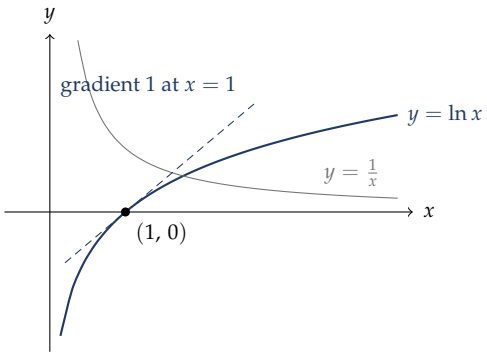
*Differentiate* $y = \ln\left(x^2 - 3x + 1\right)$.

SYLLABUS MAV-12-04

- Use chain rule on the identity $e^{\ln x} = x$ to prove the formula $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ for the derivative of the natural logarithm function where $x > 0$
- Use the chain rule to differentiate functions of the form $\ln f(x)$

NOTES

- $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$, for $x > 0$.
- Proof: differentiate both sides of $e^{\ln x} = x$. Then $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$, so $x\,\dfrac{d}{dx}(\ln x) = 1$.
- $\dfrac{d}{dx}\left(\ln f(x)\right) = \dfrac{f'(x)}{f(x)}$: the derivative of the inside, over the inside. This is on the reference sheet.
- $\ln(kx) = \ln k + \ln x$, so $\ln(kx)$ has the same derivative as $\ln x$: $\dfrac{1}{x}$.
- A product, quotient or power inside the log: expand with the log laws first. $\ln(AB) = \ln A + \ln B$, $\ln\dfrac{A}{B} = \ln A - \ln B$, $\ln A^n = n \ln A$.
- A sum or difference inside the log: differentiate it as it is.




WORKED EXAMPLE

*Differentiate* $y = \ln\left(x^2 + 4x - 1\right)$.

$f(x) = x^2 + 4x - 1$, so $f'(x) = 2x + 4$. the inside and its derivative

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{2x+4}{x^2+4x-1}$$

## Tier 1 · the log of a polynomial

**153.** For each function, write the inside $f(x)$, then find $f'(x)$.

a) $\ln(9x)$

b) $\ln\left(x^2-7\right)$

c) $\ln(5-2x)$

**154.** Expand using the log laws.

a) $\ln\left(x^3(x+2)\right)$

b) $\ln\left(\dfrac{x-1}{x+5}\right)$

c) $\ln\left((4x+3)^2\right)$

**155.** Should you use the log laws first? Answer yes or no.

a) $\ln\left((x+1)(x+2)\right)$

b) $\ln\left(2x^3+1\right)$

c) $\ln\left((x-3)^4\right)$

**156.** Differentiate:

a) $y=\ln\left(x^2+1\right)$

b) $y=\ln\left(x^3-4x\right)$

**157.** Differentiate $y=\ln\left(2x^2-5x+3\right)$.

## Tier 2 · a linear inside

**158.** Differentiate $y=\ln(6x)$.

**159.** Differentiate $y=\ln(3x+2)$.

**160.** Differentiate $y=\ln(1-4x)$.

## Tier 3 · coefficients and sums

**161.** Differentiate $y=5\ln(2x+3)$.

**162.** Differentiate $y=x^3-2\ln x$.

**163.** Differentiate $y=\ln\left(x^2+1\right)+e^{2x}$.

### Tier 4 · the power law first

**164.** Differentiate $y = \ln\left((2x+1)^3\right)$.

**165.** Differentiate $y = \ln\sqrt{4x-1}$.

**166.** Differentiate $y = \ln\left(x^5\right)$.

### Tier 5 · the product and quotient laws first

**167.** Differentiate $y = \ln\left(x(x+3)\right)$, giving your answer as a single fraction.

**168.** Differentiate $y = \ln\left(\dfrac{x+2}{x-1}\right)$, giving your answer as a single fraction.

**169.** Differentiate $y = \ln\left(\dfrac{3x-1}{x+1}\right)$, giving your answer as a single fraction.

### Tier 6 · several log laws

**170.** Differentiate $y = \ln\left(x^3\sqrt{x+1}\right)$, giving your answer as a single fraction.

**171.** Differentiate $y = \ln\left(\dfrac{(x+1)^2}{x-2}\right)$, giving your answer as a single fraction.

**172.** Differentiate $y = \ln\sqrt{\dfrac{x+1}{x-1}}$, simplifying fully.

### Conceptual

**173.** Use the identity $e^{\ln x} = x$ to prove that $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

**174.** Show that $\ln(5x)$ and $\ln x$ have the same derivative, and explain why using the log laws.

**175.** A student writes $\dfrac{d}{dx}\ln\left(x^2+1\right) = \dfrac{1}{x^2+1}$. Explain the error and give the correct derivative.

**176.** Find the exact gradient of the tangent to $y = \ln\left(x^2+3\right)$ at the point where $x = 1$.

**177.** Let $f(x) = \ln\left(x^2 - 2x + 5\right)$.

(a) Find $f'(x)$. (1 mark)

(b) Explain why $f(x)$ is defined for all real $x$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

## Revision

**178.** Differentiate $y = e^{3x} + e^{-x}$.

**179.** Differentiate $y = x^2 e^x$, giving your answer in factorised form.

**180.** Differentiate $y = 3 \cdot 5^x$.

**181.** Find the equation of the tangent to $y = e^x$ at the point where $x = 0$.

**182.** Find the equation of the normal to $y = e^x$ at the point where $x = 0$.

**183.** An investment grows according to $A = 100e^{0.1t}$. Find $\dfrac{dA}{dt}$ when $t = 0$.

**184.** Evaluate $e^{2\ln 3}$.

**185.** Solve $\ln x = 2$.

**186.** Solve $\ln(x + 1) = 0$.

**187.** Write $2\ln 3 + \ln 4$ as a single logarithm.

**188.** Solve $\log_2 x = 5$.

**189.** Write $\log_5 20$ in terms of natural logarithms.

**190.** Differentiate $y = \left(3x^2 - 1\right)^5$.

**191.** Differentiate $y = x\sqrt{x}$.

**192.** Find the value of $x$ where the curve $y = x^2 - 6x$ has gradient 4.

**193.** Solve $x^2 - 7x + 12 = 0$.

**194.** Use the discriminant to state how many real roots $x^2 + x + 3 = 0$ has.

**195.** Find the exact value of $\tan \dfrac{\pi}{3}$.

**196.** Solve $\sin x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

**197.** State the domain of $y = \sqrt{4 - x}$.

**198.** State the range of $y = \ln x$.

**199.** Solve $3x - 2 > 7$.

**200.** If $f(x) = 2x - 1$, find $f^{-1}(5)$.

**201.** Events $A$ and $B$ are independent, with $P(A) = 0.5$ and $P(B) = 0.4$. Find $P(A \cup B)$.

**202.** Simplify $\dfrac{x^2 - 4x}{x^2 - 16}$.

# 5 $\log_a x$, combined rules and tangents

ANCHOR QUESTION

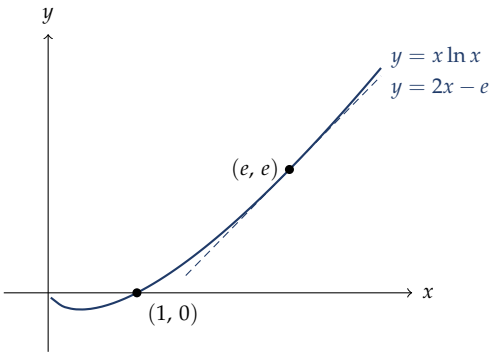
*Differentiate $y = x^2 \ln x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Prove and use the formula $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$, where $a$ is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Change of base: $\log_a x = \dfrac{\ln x}{\ln a}$.
- $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$. Proof: $\log_a x = \dfrac{1}{\ln a} \cdot \ln x$, and $\dfrac{1}{\ln a}$ is a constant.
- For $\log_a f(x)$: change to $\dfrac{\ln f(x)}{\ln a}$, then use $\dfrac{f'(x)}{f(x)}$.
- A power inside the log, as in $\ln(x^3)$: log law first, $3 \ln x$.
- A power on the whole log, as in $(\ln x)^3$: chain rule on the outside, $3(\ln x)^2 \times \dfrac{1}{x}$. Note $\ln^n x$ means $(\ln x)^n$.
- Tangents and normals at $x = e$: use $\ln e = 1$ and $\ln 1 = 0$ to keep answers exact.




WORKED EXAMPLE

*Differentiate $y = x^4 \ln x$, giving your answer in factorised form.*

$u = x^4$, $u' = 4x^3$; $\quad v = \ln x$, $v' = \dfrac{1}{x}$ — name the parts

$\dfrac{dy}{dx} = 4x^3 \ln x + x^4 \times \dfrac{1}{x} = 4x^3 \ln x + x^3$ — product rule

$= x^3(4 \ln x + 1)$ — common factor $x^3$

## Tier 1 · products with $\ln x$

**203.** Write using natural logarithms.

a) $\log_6 x$

b) $\log_3 \left(x^2 - 1\right)$

**204.** Differentiate:

a) $y = \log_4 x$

b) $y = 3\log_5 x$

c) $y = \log_2(3x - 1)$

**205.** For each function, state whether to use a log law first or the chain rule on the outside.

a) $\ln\left(x^6\right)$

b) $(\ln x)^6$

c) $\ln^2(3x)$

d) $\ln\left((x+4)^3\right)$

**206.** Differentiate $y = x\ln x$.

**207.** Differentiate $y = x^5 \ln x$, giving your answer in factorised form.

## Tier 2 · other products with logs

**208.** Differentiate $y = e^x \ln x$.

**209.** Differentiate $y = x^2 \ln(3x)$, giving your answer in factorised form.

**210.** Differentiate $y = \sqrt{x}\,\ln x$, giving your answer as a single fraction.

## Tier 3 · the quotient rule with logs

**211.** Differentiate $y = \dfrac{\ln x}{x^2}$.

**212.** Differentiate $y = \dfrac{x}{\ln x}$.

**213.** Differentiate $y = \dfrac{\ln x}{x+1}$, giving your answer as a single fraction.

## Tier 4 · powers of $\ln x$

**214.** Differentiate $y = (\ln x)^2$.

**215.** Differentiate $y = (\ln x)^4$.

**216.** Differentiate $y = \sqrt{\ln x}$.

## Tier 5 · tangents with logs

**217.** Find the equation of the tangent to $y = \ln x$ at the point where $x = 1$.

**218.** Find the equation of the tangent to $y = \ln(3x)$ at the point where $x = \frac{1}{3}$.

**219.** Find the equation of the tangent to $y = 2\ln x$ at the point where $x = e$.

## Tier 6 · normals and combined rules

**220.** Find the equation of the normal to $y = \ln x$ at the point where $x = e$.

**221.** Find the equation of the tangent to $y = x^2 \ln x$ at the point where $x = 1$.

**222.** Find the equation of the normal to $y = x\ln x$ at the point where $x = 1$.

## Conceptual

**223.** Prove that $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x\ln a}$.

**224.** Find $\dfrac{d}{dx}\ln(x^2)$ and $\dfrac{d}{dx}(\ln x)^2$, and explain why they are different.

**225.** Show that the tangent to $y = \ln x$ at the point where $x = e$ passes through the origin.

**226.** Find the exact coordinates of the point on $y = x\ln x$ where the tangent is horizontal.

**227.** Let $f(x) = \dfrac{\ln x}{x}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**228.** Differentiate $y = \ln\left(x^2 + 7\right)$.

**229.** Differentiate $y = \ln\left(\dfrac{x+1}{x-1}\right)$, giving your answer as a single fraction.

**230.** Differentiate $y = xe^{-x}$, giving your answer in factorised form.

**231.** Differentiate $y = 10^x$.

**232.** Differentiate $y = e^{\sqrt{x}}$.

**233.** Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 0$.

**234.** Show that $\dfrac{d}{dx}\left(e^{2x} - e^{-2x}\right) = 2\left(e^{2x} + e^{-2x}\right)$.

**235.** Solve $e^{2x} - 3e^x + 2 = 0$.

**236.** Solve $\ln(2x - 1) = 0$.

**237.** Simplify $\ln\left(e^x\right) + e^{\ln x}$.

**238.** Evaluate $\log_4 2$.

**239.** Write $\ln 18 - \ln 2 + \ln 5$ as a single logarithm.

**240.** Differentiate $y = \dfrac{4x+1}{2x-3}$.

**241.** Differentiate $y = \dfrac{6}{\sqrt{x}}$.

**242.** Find the equation of the normal to $y = x^2$ at the point $(1, 1)$. Give your answer in general form.

**243.** Solve $x^3 - 4x = 0$.

**244.** Find the values of $k$ for which $x^2 + kx + 9 = 0$ has exactly one real root.

**245.** Find the exact value of $\cos\dfrac{5\pi}{6}$.

**246.** Solve $2\sin^2 x = 1$ for $0 \leq x \leq 2\pi$.

**247.** State the period of $y = \sin(3x)$.

**248.** State the domain of $y = \dfrac{1}{x-2}$.

**249.** Solve $|x + 1| \geq 3$.

**250.** If $f(x) = \sqrt{x}$ and $g(x) = x^2 + 9$, find $f(g(4))$.

**251.** If $P(A) = 0.3$ and $P(B \mid A) = 0.5$, find $P(A \cap B)$.

**252.** Factorise $x^3 - 8$.

# 6 Derivatives of $\sin x$, $\cos x$ and $\tan x$

ANCHOR QUESTION

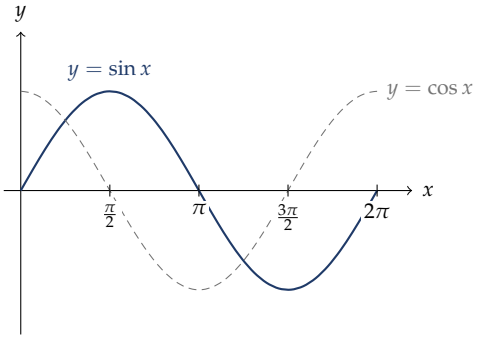
*Differentiate* $y = \sin\left(x^2 + 2x\right)$.

SYLLABUS MAV-12-04

- Determine the formulas $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ informally by using the graphs of $y = \sin x$ and $y = \cos x$ and verify with graphing applications
- Use the rules of differentiation to show that $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the chain rule to differentiate functions of the form $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$

NOTES

- $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$. Angles must be in radians.
- From the graph of $y = \sin x$: the gradient is 0 at the turning points $x = \frac{\pi}{2}$, $\frac{3\pi}{2}$, is 1 at $x = 0$ and $-1$ at $x = \pi$. That is the graph of $y = \cos x$.
- $\frac{d}{dx}(\tan x) = \sec^2 x$, where $\sec x = \frac{1}{\cos x}$. Proof: write $\tan x = \frac{\sin x}{\cos x}$ and use the quotient rule with $\sin^2 x + \cos^2 x = 1$.
- Chain rule: $\frac{d}{dx}\sin f(x) = f'(x)\cos f(x)$, $\frac{d}{dx}\cos f(x) = -f'(x)\sin f(x)$, $\frac{d}{dx}\tan f(x) = f'(x)\sec^2 f(x)$.
- The angle stays the same; multiply by the derivative of the angle.




WORKED EXAMPLE

*Differentiate* $y = \cos\left(x^2 + 4x\right)$.

$f(x) = x^2 + 4x$, so $f'(x) = 2x + 4$. — the angle and its derivative

$\frac{dy}{dx} = -f'(x)\sin f(x)$ — cos gives − sin

$= -(2x + 4)\sin\left(x^2 + 4x\right)$

## Tier 1 · $\sin f(x)$

**253.** For each function, write the angle $f(x)$, then find $f'(x)$.

a) $\sin(9x)$

b) $\cos\left(x^2 - 1\right)$

c) $\tan(5 - 2x)$

**254.** Differentiate:

a) $y = \sin x + \cos x$

b) $y = 4\tan x$

c) $y = x^2 - 3\cos x$

**255.** Differentiate $y = \sin\left(x^2 + 3x\right)$.

**256.** Differentiate $y = \sin\left(x^3\right)$.

**257.** Differentiate $y = \sin(5x - 2)$.

## Tier 2 · $\cos f(x)$

**258.** Differentiate $y = \cos\left(x^2 + 1\right)$.

**259.** Differentiate $y = \cos 4x$.

**260.** Differentiate $y = \cos(2 - 3x)$.

## Tier 3 · $\tan f(x)$

**261.** Differentiate $y = \tan 6x$.

**262.** Differentiate $y = \tan\left(x^2\right)$.

**263.** Differentiate $y = \tan(1 - x)$.

## Tier 4 · coefficients and sums

**264.** Differentiate $y = 5\sin 3x + 2\cos 4x$.

**265.** Differentiate $y = 2\tan\dfrac{x}{2}$.

**266.** Differentiate $y = \cos(\pi x) - \sin(\pi x)$.

## Tier 5 · exact gradients

**267.** Find the exact gradient of the tangent to $y = \cos 2x$ at the point where $x = \dfrac{\pi}{12}$.

**268.** Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{4}$.

**269.** Find the exact gradient of the tangent to $y = \sin 3x$ at the point where $x = \dfrac{\pi}{9}$.

## Tier 6 · horizontal tangents

**270.** Find the values of $x$, for $0 \leq x \leq 2\pi$, where the tangent to $y = \sin x - \cos x$ is horizontal.

**271.** Find the values of $x$, for $0 \leq x \leq 2\pi$, where the tangent to $y = x + 2\cos x$ is horizontal.

**272.** Find the values of $x$, for $0 \leq x \leq \pi$, where the tangent to $y = \sin 2x$ is horizontal.

## Conceptual

**273.** Use the quotient rule to show that $\dfrac{d}{dx}(\tan x) = \sec^2 x$.

**274.** Use the graph of $y = \sin x$ to explain why its gradient is 1 at $x = 0$, 0 at $x = \dfrac{\pi}{2}$ and $-1$ at $x = \pi$, and why this matches $y = \cos x$.

**275.** A student writes $\dfrac{d}{dx}(\cos 3x) = -\sin 3x$. Explain the error and give the correct derivative.

**276.** Explain why the graph of $y = \tan x$ has no horizontal tangents.

**277.** Let $f(x) = \sin x + \sqrt{3}\cos x$, for $0 \leq x \leq 2\pi$.

(a) Find $f'(x)$. (1 mark)

(b) Find the values of $x$ where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**278.** Differentiate $y = \ln(5x - 2)$.

**279.** Differentiate $y = x \ln x$.

**280.** Differentiate $y = \log_3 x$.

**281.** Differentiate $y = (\ln x)^2$.

**282.** Differentiate $y = e^{-3x}$.

**283.** Differentiate $y = x^3 e^{2x}$, giving your answer in factorised form.

**284.** Find the equation of the tangent to $y = \ln x$ at the point where $x = e^2$.

**285.** Convert $225^\circ$ to radians.

**286.** Convert $\dfrac{7\pi}{6}$ radians to degrees.

**287.** Find the exact value of $\sin \dfrac{5\pi}{4}$.

**288.** Find the exact value of $\tan \dfrac{2\pi}{3}$.

**289.** Solve $\cos x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

**290.** Solve $\tan x = \sqrt{3}$ for $0 \leq x \leq 2\pi$.

**291.** Simplify $\dfrac{1 - \cos^2 x}{\sin x}$.

**292.** State the amplitude and period of $y = 3\cos 2x$.

**293.** Find the exact length of an arc of a circle with radius 6 cm that subtends an angle of $\dfrac{\pi}{3}$ at the centre.

**294.** Find the exact area of a sector with radius 4 cm and angle $\dfrac{\pi}{2}$.

**295.** Solve $2^{x+1} = 32$.

**296.** Solve $\ln x + \ln 2 = \ln 10$.

**297.** Differentiate $y = \left(x^2 - 3\right)^4$.

**298.** Find the equation of the normal to $y = x^3$ at the point $(1, 1)$. Give your answer in general form.

**299.** Solve $x^2 + 4x - 5 < 0$.

**300.** Is $f(x) = \sin x$ even, odd or neither? Give a reason.

**301.** Two fair dice are rolled. Find the probability that both show the same number.

**302.** Expand $(2x - 1)(x + 3)$.

# 7 Trig functions with combined rules

ANCHOR QUESTION

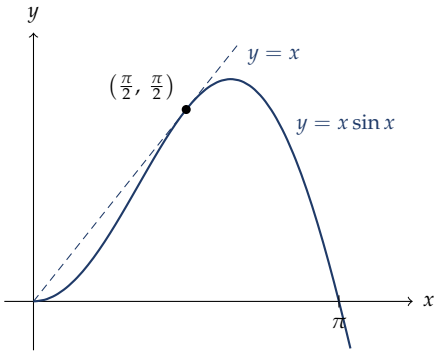
*Differentiate $y = x^2 \sin x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sin^n x$ means $(\sin x)^n$: use the chain rule on the outside, e.g. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$.
- Product rule $u'v + uv'$ and quotient rule $\dfrac{u'v - uv'}{v^2}$ work with trig functions exactly as before. Name $u$, $v$, $u'$, $v'$ first.
- Take out common factors, e.g. $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$.
- Trig inside $e$ or ln: $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$ and $\dfrac{d}{dx}\ln(\cos x) = \dfrac{-\sin x}{\cos x} = -\tan x$.
- For tangents, use exact values: $\sin\dfrac{\pi}{2} = 1$, $\cos\dfrac{\pi}{2} = 0$, $\cos\pi = -1$.




WORKED EXAMPLE

*Differentiate $y = x^3\cos x$, giving your answer in factorised form.*

$u = x^3$, $u' = 3x^2$; $\quad v = \cos x$, $v' = -\sin x$ — name the parts

$\dfrac{dy}{dx} = 3x^2\cos x - x^3\sin x$ — product rule

$= x^2(3\cos x - x\sin x)$ — common factor $x^2$

## Tier 1 · products with $\sin x$, $\cos x$, $\tan x$

**303.** Rewrite with the power outside a bracket.

a) $\cos^3 x$

b) $\sin^2 4x$

**304.** Differentiate $y = x \sin x$.

**305.** Differentiate $y = x^2 \cos x$, giving your answer in factorised form.

**306.** Differentiate $y = x^3 \tan x$, giving your answer in factorised form.

**307.** Differentiate $y = \sqrt{x}\, \sin x$.

## Tier 2 · products with a chain inside

**308.** Differentiate $y = x \sin 3x$.

**309.** Differentiate $y = x^2 \cos 2x$, giving your answer in factorised form.

**310.** Differentiate $y = \sin x \cos x$.

## Tier 3 · powers of trig functions

**311.** Differentiate $y = \cos^2 x$.

**312.** Differentiate $y = \sin^3 x$.

**313.** Differentiate $y = \tan^2 x$.

## Tier 4 · the quotient rule with trig

**314.** Differentiate $y = \dfrac{\cos x}{x}$.

**315.** Differentiate $y = \dfrac{x}{\sin x}$.

**316.** Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{1+\cos x}\right) = \dfrac{1}{1+\cos x}$.

## Tier 5 · trig with $e$ and $\ln$

**317.** Differentiate $y = e^x \sin x$, giving your answer in factorised form.

**318.** Differentiate $y = \ln(\sin x)$.

**319.** Differentiate $y = e^{\cos x}$.

## Tier 6 · tangents and normals

**320.** Find the equation of the tangent to $y = x\cos x$ at the point where $x = \pi$.

**321.** Find the equation of the tangent to $y = \sin^2 x$ at the point where $x = \dfrac{\pi}{4}$.

**322.** Find the equation of the normal to $y = e^x \cos x$ at the point where $x = 0$.

## Tier 7 · show that

**323.** Show that $\dfrac{d}{dx}\left(\dfrac{\cos x}{1-\sin x}\right) = \dfrac{1}{1-\sin x}$.

**324.** Show that $\dfrac{d}{dx}(x\sin x + \cos x) = x\cos x$.

**325.** Show that $\dfrac{d}{dx}\left[e^{-x}(\sin x + \cos x)\right] = -2e^{-x}\sin x$.

## Conceptual

**326.** Show that $\dfrac{d}{dx}\left(\sin^2 x + \cos^2 x\right) = 0$, and explain why this makes sense.

**327.** Show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$.

**328.** A student writes $\dfrac{d}{dx}(x\sin x) = \cos x$. Explain the error and give the correct derivative.

**329.** Find the exact gradient of the tangent to $y = x^2\sin x$ at the point where $x = \pi$.

**330.** Let $f(x) = e^{-x}\sin x$, for $0 \le x \le 2\pi$.

(a) Show that $f'(x) = e^{-x}(\cos x - \sin x)$. (1 mark)

(b) Find the values of $x$ where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**331.** Differentiate $y = \sin\left(x^2\right)$.

**332.** Differentiate $y = \tan 5x$.

**333.** Differentiate $y = \cos(3x + 1)$.

**334.** Find the exact gradient of the tangent to $y = \cos x$ at the point where $x = \dfrac{\pi}{6}$.

**335.** Differentiate $y = e^{x^2}$.

**336.** Differentiate $y = \ln\left(x^3 + 1\right)$.

**337.** Differentiate $y = 2^{3x}$.

**338.** Differentiate $y = \log_{10} x$.

**339.** Differentiate $y = xe^x$, giving your answer in factorised form.

**340.** Differentiate $y = \dfrac{\ln x}{x}$.

**341.** Find the equation of the tangent to $y = e^x$ at the point where $x = 1$.

**342.** Find the equation of the normal to $y = \ln x$ at the point where $x = 1$.

**343.** Find the exact value of $\cos \dfrac{7\pi}{6}$.

**344.** Solve $\sin 2x = 0$ for $0 \leq x \leq \pi$.

**345.** Solve $2\cos x + \sqrt{3} = 0$ for $0 \leq x \leq 2\pi$.

**346.** Simplify $\tan^2 x + 1$.

**347.** State the range of $y = 2\sin x + 1$.

**348.** State the period of $y = \tan 2x$.

**349.** Solve $e^x = \dfrac{1}{e}$.

**350.** Simplify $\ln\left(e^2 x\right) - \ln x$.

**351.** Evaluate $\log_2 48 - \log_2 3$.

**352.** Differentiate $y = (1 - 2x)^5$.

**353.** Differentiate $y = \dfrac{x^2}{x + 1}$.

**354.** Solve $x^2 - 2x - 1 = 0$, giving exact answers.

**355.** Find the gradient of a line perpendicular to $3x + 4y = 12$.

**356.** If $f(x) = x^2 - 1$, find and simplify $f(a + 1)$.

**357.** A fair coin is tossed three times. Find the probability of at least one head.

**358.** Simplify $\sqrt{50} - \sqrt{18}$.

# 8 Derivatives of the reciprocal trig functions

ANCHOR QUESTION

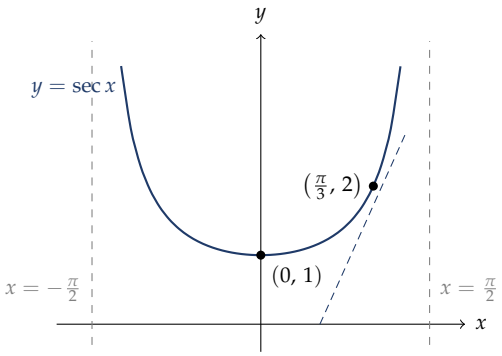
*Show that* $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

SYLLABUS MAV-12-04

- Use the rules of differentiation to find the derivatives of cosec $x$, sec $x$ and cot $x$
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sec x = \dfrac{1}{\cos x} = (\cos x)^{-1}$, $\operatorname{cosec} x = \dfrac{1}{\sin x} = (\sin x)^{-1}$, $\cot x = \dfrac{\cos x}{\sin x}$.
- $\dfrac{d}{dx}(\sec x) = \tan x \sec x$: chain rule on $(\cos x)^{-1}$.
- $\dfrac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$: chain rule on $(\sin x)^{-1}$.
- $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$: quotient rule on $\dfrac{\cos x}{\sin x}$.
- These three are *not* on the reference sheet. The "co" functions get the minus sign.
- With an angle $f(x)$, multiply by $f'(x)$ as usual, e.g. $\dfrac{d}{dx}\sec 2x = 2\tan 2x \sec 2x$.




WORKED EXAMPLE

*Show that* $\dfrac{d}{dx}(\sec 4x) = 4\tan 4x \sec 4x$.

$y = (\cos 4x)^{-1}$ — write as a power

$\dfrac{dy}{dx} = -(\cos 4x)^{-2} \times (-4\sin 4x)$ — chain rule

$= \dfrac{4\sin 4x}{\cos^2 4x} = 4 \cdot \dfrac{\sin 4x}{\cos 4x} \cdot \dfrac{1}{\cos 4x} = 4\tan 4x \sec 4x$

## Tier 1 · $\sec x$ as a power

**359.** Write as a power of a trig function.

a) $\operatorname{cosec} 5x$

b) $\sec\left(x^3\right)$

**360.** Find the exact value.

a) $\sec \dfrac{\pi}{6}$

b) $\operatorname{cosec} \dfrac{\pi}{3}$

c) $\cot \dfrac{\pi}{4}$

d) $\operatorname{cosec} \dfrac{\pi}{2}$

**361.** By writing $\sec 3x = (\cos 3x)^{-1}$, show that $\dfrac{d}{dx}(\sec 3x) = 3\tan 3x \sec 3x$.

**362.** Differentiate $y = \sec^2 x$.

**363.** Differentiate $y = 5\sec x$.

## Tier 2 · $\operatorname{cosec} x$

**364.** Differentiate $y = \operatorname{cosec} x + x^2$.

**365.** Differentiate $y = 2\operatorname{cosec} x$.

**366.** Show that $\dfrac{d}{dx}(\operatorname{cosec} 2x) = -2\cot 2x \operatorname{cosec} 2x$.

## Tier 3 · $\cot x$

**367.** Differentiate $y = \cot x - x$.

**368.** Differentiate $y = 4\cot x$.

**369.** By writing $\cot x = (\tan x)^{-1}$, show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

## Tier 4 · sums

**370.** Differentiate $y = \sec x + \operatorname{cosec} x$.

**371.** Differentiate $y = 2\cot x - 3\sec x$.

**372.** Differentiate $y = \tan x + \cot x$.

## Tier 5 · the chain rule

**373.** Differentiate $y = \sec\left(x^2\right)$.

**374.** Differentiate $y = \cot 3x$.

**375.** Differentiate $y = \operatorname{cosec}(2x - 1)$.

## Tier 6 · the product rule

**376.** Differentiate $y = x \operatorname{cosec} x$, giving your answer in factorised form.

**377.** Differentiate $y = x^2 \cot x$, giving your answer in factorised form.

**378.** Differentiate $y = e^x \sec x$, giving your answer in factorised form.

## Tier 7 · tangents and normals

**379.** Find the equation of the tangent to $y = \sec x$ at the point where $x = \dfrac{\pi}{3}$.

**380.** Find the gradient of the normal to $y = \operatorname{cosec} x$ at the point where $x = \dfrac{\pi}{4}$.

**381.** Find the equation of the tangent to $y = \cot x$ at the point where $x = \dfrac{\pi}{2}$.

## Conceptual

**382.** Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

**383.** Use the quotient rule to show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

**384.** Explain why $y = \cot x$ is decreasing wherever it is defined.

**385.** A student says that since $\sec x = \dfrac{1}{\cos x}$, its derivative is $\dfrac{1}{-\sin x}$. Explain the error.

**386.** Let $f(x) = \sec x$, for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

(a) Show that $f'(x) = \dfrac{\sin x}{\cos^2 x}$. (1 mark)

(b) Find the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

(c) Explain why $f'(x) > 0$ for $0 < x < \dfrac{\pi}{2}$. (1 mark)

## Revision

**387.** Differentiate $y = x \sin x$.

**388.** Differentiate $y = \cos^2 x$.

**389.** Differentiate $y = \tan 3x$.

**390.** Differentiate $y = e^{\sin x}$.

**391.** Differentiate $y = \dfrac{\sin x}{x}$.

**392.** Differentiate $y = \ln\left(x^2 + 1\right)$.

**393.** Differentiate $y = 5^x$.

**394.** Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

**395.** Find the equation of the tangent to $y = \sin x$ at the point where $x = \pi$.

**396.** Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{3}$.

**397.** Solve $\sin x = \cos x$ for $0 \leq x \leq 2\pi$.

**398.** Find the exact value of $\sin \dfrac{4\pi}{3}$.

**399.** Convert $300^\circ$ to radians.

**400.** Simplify $\sin x \operatorname{cosec} x$.

**401.** Simplify $1 + \cot^2 x$.

**402.** Solve $2 \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$.

**403.** Solve $\ln(x - 2) = \ln 5$.

**404.** Solve $3e^{2x} = 12$, giving the exact answer.

**405.** Evaluate $\log_3 5 + \log_3 \frac{1}{5}$.

**406.** Differentiate $y = 3x^4 - \dfrac{2}{x}$.

**407.** Differentiate $y = \sqrt{x^2 + 4}$.

**408.** Find the values of $x$ where the curve $y = x^3 - 12x$ has a horizontal tangent.

**409.** Solve $x^2 > 9$.

**410.** The line $y = 2x + c$ passes through $(3, 1)$. Find $c$.

**411.** Solve simultaneously: $y = x^2$ and $y = x + 2$.

**412.** State the domain of $y = \ln\left(x^2\right)$.

**413.** Two fair dice are rolled. Find the probability that the sum is 10.

**414.** Simplify $\left(3a^2b\right)^3$.

# 9 Mixed differentiation: choosing the rule

ANCHOR QUESTION

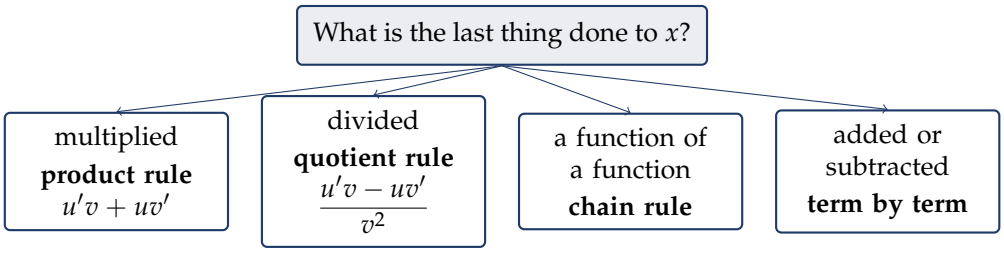
*Differentiate $y = e^{2x}\sin 3x$, giving your answer in factorised form.*

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Standard derivatives: $e^x \to e^x$, $\ln x \to \dfrac{1}{x}$, $a^x \to (\ln a)\,a^x$, $\sin x \to \cos x$, $\cos x \to -\sin x$, $\tan x \to \sec^2 x$, $\sec x \to \tan x \sec x$, $\operatorname{cosec} x \to -\cot x \operatorname{cosec} x$, $\cot x \to -\operatorname{cosec}^2 x$.
- Choose the rule from the *last* thing done to $x$: multiplied, divided, a function of a function, or added.
- Inside a product or quotient, each part may need its own chain rule, e.g. $\dfrac{d}{dx}\sin 3x = 3\cos 3x$.
- Simplify by taking out common factors, especially exponentials.
- $e^{f(x)}$, $\ln f(x)$, $\sin f(x)$ and $[f(x)]^n$ all follow the chain rule: keep the inside, multiply by its derivative.




WORKED EXAMPLE

*Differentiate $y = e^{-x}\sin 2x$, giving your answer in factorised form.*

Last operation: multiplied, so the product rule. — choose the rule

$u = e^{-x}$, $u' = -e^{-x}$; $\quad v = \sin 2x$, $v' = 2\cos 2x$ — chain rule on each part

$$\frac{dy}{dx} = -e^{-x}\sin 2x + 2e^{-x}\cos 2x$$

$$= e^{-x}(2\cos 2x - \sin 2x)$$ — common factor $e^{-x}$

## Tier 1 · products with a chain inside

**415.** Differentiate:

a) $e^x$

b) $\ln x$

c) $\sin x$

d) $\cot x$

e) $3^x$

**416.** Which rule comes first?

a) $x^2 \cos x$

b) $\cos\left(x^2\right)$

c) $\dfrac{\cos x}{x^2}$

d) $x^2 + \cos x$

**417.** Differentiate $y = e^x \cos 2x$, giving your answer in factorised form.

**418.** Differentiate $y = x^2 e^{3x}$, giving your answer in factorised form.

## Tier 2 · products with a negative power

**419.** Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

**420.** Differentiate $y = xe^{-3x}$, giving your answer in factorised form.

**421.** Differentiate $y = e^{-x} \cos x$, giving your answer in factorised form.

## Tier 3 · quotients

**422.** Differentiate $y = \dfrac{\ln x}{e^x}$, giving your answer as a single fraction.

**423.** Differentiate $y = \dfrac{\sin x}{e^{2x}}$.

**424.** Differentiate $y = \dfrac{x^2}{\ln x}$, giving your answer in factorised form.

## Tier 4 · a function of a function

**425.** Differentiate $y = \cos\left(e^x\right)$.

**426.** Differentiate $y = e^{\tan x}$.

**427.** Differentiate $y = \ln\left(1 + e^x\right)$.

### Tier 5 · the log of a sum

**428.** Differentiate $y = \ln(x + \cos x)$.

**429.** Differentiate $y = \ln\left(e^{x} + e^{-x}\right)$.

**430.** Differentiate $y = \ln(2x + \sin 2x)$.

### Tier 6 · powers of functions

**431.** Differentiate $y = (e^{x} + 1)^{3}$.

**432.** Differentiate $y = (\sin x + \cos x)^{2}$.

**433.** Differentiate $y = \sqrt{1 + \ln x}$.

### Tier 7 · quotients with a chain

**434.** Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{(1+\cos x)^{2}}\right) = \dfrac{2 - \cos x}{(1+\cos x)^{2}}$.

**435.** Differentiate $y = \dfrac{x}{\left(x^{2}+1\right)^{2}}$, simplifying fully.

**436.** Differentiate $y = \dfrac{e^{x}}{\left(e^{x}+1\right)^{3}}$, simplifying fully.

### Tier 8 · tangents and normals

**437.** Find the equation of the tangent to $y = e^{x}\sin x$ at the origin.

**438.** Find the equation of the tangent to $y = \ln\left(1 + x^{2}\right)$ at the point where $x = 1$.

**439.** Find the equation of the normal to $y = xe^{2x}$ at the origin.

### Conceptual

**440.** Explain why $y = e^{x}\sin x$ needs the product rule but $y = e^{\sin x}$ needs the chain rule, and differentiate both.

**441.** Show that $\dfrac{d}{dx}\ln(\sec x) = \tan x$.

**442.** A student writes $\dfrac{d}{dx}e^{x^{2}} = x^{2}e^{x^{2}-1}$. Explain the error and give the correct derivative.

**443.** Find the exact gradient of the tangent to $y = x\ln x - x$ at the point where $x = e$.

**444.** Let $f(x) = \dfrac{\ln x}{x^2}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - 2\ln x}{x^3}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

## Revision

**445.** Differentiate $y = e^{5x-1}$.

**446.** Differentiate $y = \ln\left(3x^2\right)$.

**447.** Differentiate $y = \log_2(x+1)$.

**448.** Differentiate $y = 7^x$.

**449.** Differentiate $y = \tan\left(x^2\right)$.

**450.** Differentiate $y = \sec 3x$.

**451.** Differentiate $y = \operatorname{cosec} x$.

**452.** Differentiate $y = \cot 2x$.

**453.** Differentiate $y = \sin^2 3x$.

**454.** Differentiate $y = x\cos x$.

**455.** Find the equation of the tangent to $y = e^{-x}$ at the point where $x = 0$.

**456.** Find the equation of the normal to $y = \sin x$ at the origin.

**457.** Show that $\dfrac{d}{dx}\left(\dfrac{e^x - 1}{e^x + 1}\right) = \dfrac{2e^x}{(e^x+1)^2}$.

**458.** Find the exact value of $\cos\dfrac{3\pi}{4}$.

**459.** Find the exact value of $\sec\dfrac{\pi}{4}$.

**460.** Solve $\tan x = -1$ for $0 \le x \le 2\pi$.

**461.** Solve $\cos 2x = 0$ for $0 \le x \le \pi$.

**462.** Solve $e^x = 3$, giving the exact answer.

**463.** Solve $\log_5 x = 2$.

**464.** Simplify $\ln\left(\dfrac{e^3}{e}\right)$.

**465.** Differentiate $y = (3x+2)^{-2}$.

**466.** Differentiate $y = \dfrac{x^2}{1-x}$.

**467.** Find the equation of the tangent to $y = x^2 + 3x$ at the point where $x = -1$.

**468.** Solve $2x^2 - 7x + 3 = 0$.

**469.** Find the coordinates of the vertex of $y = x^2 - 6x + 5$.

**470.** Solve $|3x - 1| = 5$.

**471.** State the domain of $f(x) = \sqrt{2x - 6}$.

**472.** If $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.1$, find $P(A \cup B)$.

**473.** Find the gradient of the line through $(-1, 4)$ and $(3, -4)$.

**474.** Simplify $\left(x^{-2}\right)^3 \times x^7$.

## Solutions

**1.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = 2 - x$, $f'(x) = -1$ c) $f(x) = x^4 - 3x$, $f'(x) = 4x^3 - 3$

**2.** a) $7e^{7x}$ b) $2x\,e^{x^2+1}$ c) $(3x^2 - 6)\,e^{x^3-6x} = 3(x^2 - 2)\,e^{x^3-6x}$

**3.** a) $-5e^{1-5x}$ b) $\dfrac{1}{2\sqrt{x}}\,e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$ c) 0, since $e^4$ is a constant

**4.** a) $12e^{3x}$ b) $-10e^{5x+1}$ c) $3e^{6x}$

**5.** a) $6x\,e^{x^2}$ b) $-(4x - 1)\,e^{2x^2-x} = (1 - 4x)\,e^{2x^2-x}$

**6.** $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = \frac{1}{4} \times 8e^{8x-2} = 2e^{8x-2}$

**7.** a) $4e^{4x} - 4e^{-4x}$ b) $6e^{2x} - 3x^2$

**8.** $\dfrac{dy}{dx} = -6e^{-3x} - 5e^x + 7$

**9.** $\dfrac{dy}{dx} = 2x - 4 - e^{2-x}$

**10.** a) $y = e^{-x}$, so $\dfrac{dy}{dx} = -e^{-x}$ b) $y = 4e^{-2x}$, so $\dfrac{dy}{dx} = -8e^{-2x}$

**11.** a) $y = e^{4x}$, so $\dfrac{dy}{dx} = 4e^{4x}$ b) $y = e^{6x}$, so $\dfrac{dy}{dx} = 6e^{6x}$

**12.** a) $y = e^{2x}$, so $\dfrac{dy}{dx} = 2e^{2x}$ b) $y = e^{2x} + 2 + e^{-2x}$, so $\dfrac{dy}{dx} = 2e^{2x} - 2e^{-2x}$

**13.** $f'(x) = 2e^{2x+1}$, so $f'(0) = 2e$.

**14.** $f'(x) = 2x\,e^{x^2-4}$, so $f'(2) = 4e^0 = 4$.

**15.** $\dfrac{dy}{dx} = -3e^{-x} + e^x$. At $x = \ln 3$: $e^{\ln 3} = 3$ and $e^{-\ln 3} = \frac{1}{3}$, so the gradient is $-3 \times \frac{1}{3} + 3 = 2$.

**16.** $\dfrac{dy}{dx} = 3\left(e^x + 1\right)^2 \times e^x = 3e^x\left(e^x + 1\right)^2$

**17.** $\dfrac{dy}{dx} = 4\left(2 - e^{3x}\right)^3 \times \left(-3e^{3x}\right) = -12e^{3x}\left(2 - e^{3x}\right)^3$

**18.** $y = \left(1 + e^{2x}\right)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(1 + e^{2x}\right)^{-\frac{1}{2}} \times 2e^{2x} = \dfrac{e^{2x}}{\sqrt{1 + e^{2x}}}$

**19.** a) $y = e^{x^2-2x}$, so $\dfrac{dy}{dx} = (2x - 2)\,e^{x^2-2x} = 2(x - 1)\,e^{x^2-2x}$. b) $e^{x^2-2x} > 0$, so $x = 1$; $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**20.** a) $y = e^{2x^2-8x}$, so $\dfrac{dy}{dx} = (4x - 8)\,e^{2x^2-8x} = 4(x - 2)\,e^{2x^2-8x}$. b) $e^{2x^2-8x} > 0$, so $x = 2$; $y = e^{8-16} = e^{-8}$. The point is $\left(2,\, e^{-8}\right)$.

**21.** a) $f'(x) = (3x^2 - 3)\,e^{x^3-3x} = 3(x-1)(x+1)\,e^{x^3-3x}$ b) $x = 1$ or $x = -1$. $f(1) = e^{-2}$ and $f(-1) = e^2$, giving $(1, e^{-2})$ and $(-1, e^2)$.

**22.** The student has used the power rule, which only applies when $x$ is in the base. Here $x$ is in the power, so the power stays the same and is multiplied by its derivative: $\frac{d}{dx}\left(e^{3x}\right) = 3e^{3x}$.

**23.** Let $u = ax + b$, so $y = e^u$. Then $\frac{du}{dx} = a$ and $\frac{dy}{du} = e^u$, so $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.

**24.** $\frac{dy}{dx} = 10e^{2x}$. Since $e^{2x} > 0$ for all $x$, the gradient is always positive, so the graph is always increasing.

**25.** $\frac{dy}{dx} = (2x + k)\,e^{x^2+kx}$. The exponential is never 0, so $2(3) + k = 0$ and $k = -6$.

**26.** a) $\frac{dy}{dx} = 10e^{2x}$ b) $10e^{2x} = 2 \times 5e^{2x} = 2y$ c) $\frac{dy}{dx} = 2 \times 10 = 20$

**27.** $\frac{dy}{dx} = 4(3x-2)^3 \times 3 = 12(3x-2)^3$

**28.** $y = x^{\frac{1}{2}} + 2x^{-1}$, so $\frac{dy}{dx} = \frac{1}{2\sqrt{x}} - \frac{2}{x^2}$

**29.** $\frac{dy}{dx} = 2x(x+3)^3 + 3x^2(x+3)^2 = x(x+3)^2(5x+6)$

**30.** $\frac{dy}{dx} = \frac{2(x-3) - 1(2x+1)}{(x-3)^2} = \frac{-7}{(x-3)^2}$

**31.** $\frac{dy}{dx} = 3x^2 - 2 = 1$ at $x = 1$, and $y = -1$. So $y + 1 = 1(x-1)$, i.e. $y = x - 2$.

**32.** $\frac{dy}{dx} = 2x + 4 = 2$ at $x = -1$, so the normal has gradient $-\frac{1}{2}$.

**33.** $e^{2x-5x-x} = e^{-4x}$

**34.** $2x = \ln 9$, so $x = \frac{1}{2}\ln 9 = \ln 3$.

**35.** 5

**36.** $3 + 0 = 3$

**37.** $x + 1 = 3^2 = 9$, so $x = 8$.

**38.** $2\ln x + \ln(x+1)$

**39.** $\frac{\sqrt{3}}{2}$

**40.** $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$

**41.** 150°

**42.** $\sin^2\theta$

**43.** $x > -2$

**44.** $g(3) = 9$, so $f(9) = 19$.

**45.** $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so $f$ is odd.

**46.** $(x-1)(x-5) < 0$, so $1 < x < 5$.

**47.** $f^{-1}(x) = \dfrac{x+4}{3}$

**48.** A vertical dilation by a factor of 2 and a translation 1 unit to the right.

**49.** $0.4 \times 0.5 = 0.2$

**50.** $\dfrac{6}{36} = \dfrac{1}{6}$

**51.** $x + 3 = 2x$ gives $x = 3$. $x + 3 = -2x$ gives $x = -1$, but then $2x < 0$, so reject it. $x = 3$.

**52.** $\dfrac{dy}{dx} = 2x = 1$, so $\tan\theta = 1$ and $\theta = 45°$.

**53.** a) $e^{x\ln 9}$ b) $e^{5x\ln 2}$

**54.** a) $x^2e^{2x}(3+2x)$ b) $e^{3x}(1-3x)$

**55.** a) $(\ln 6)\,6^x$ b) $5(\ln 2)\,2^x$ c) $3^{4x} = e^{4x\ln 3}$, so $\dfrac{dy}{dx} = 4(\ln 3)\,3^{4x}$

**56.** a) $e^x + xe^x = e^x(1+x)$ b) $2xe^{4x} + 4x^2e^{4x} = 2xe^{4x}(1+2x)$

**57.** $\dfrac{dy}{dx} = 4x^3e^x + x^4e^x = x^3e^x(4+x)$

**58.** $\dfrac{dy}{dx} = 3e^{-x} - (3x-1)e^{-x} = e^{-x}(4-3x)$

**59.** $\dfrac{dy}{dx} = e^{-2x} - 2xe^{-2x} = e^{-2x}(1-2x)$

**60.** $\dfrac{dy}{dx} = 2xe^{-3x} - 3x^2e^{-3x} = xe^{-3x}(2-3x)$

**61.** $\dfrac{dy}{dx} = \dfrac{3e^{3x}\cdot x - e^{3x}\cdot 1}{x^2} = \dfrac{e^{3x}(3x-1)}{x^2}$

**62.** $\dfrac{dy}{dx} = \dfrac{1\cdot e^{2x} - x\cdot 2e^{2x}}{(e^{2x})^2} = \dfrac{e^{2x}(1-2x)}{e^{4x}} = \dfrac{1-2x}{e^{2x}}$

**63.** $\dfrac{dy}{dx} = \dfrac{e^x(x+1) - e^x\cdot 1}{(x+1)^2} = \dfrac{xe^x}{(x+1)^2}$

**64.** $\dfrac{dy}{dx} = 2^x + x(\ln 2)\,2^x = 2^x(1 + x\ln 2)$

**65.** $\dfrac{dy}{dx} = 2x \cdot 5^x + x^2(\ln 5)\, 5^x = x \cdot 5^x(2 + x \ln 5)$

**66.** $\dfrac{dy}{dx} = (\ln 2)\, 2^x e^x + 2^x e^x = 2^x e^x(\ln 2 + 1)$

**67.** $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 2) - e^x \cdot e^x}{(e^x + 2)^2} = \dfrac{2e^x}{(e^x + 2)^2}$

**68.** $\dfrac{dy}{dx} = \dfrac{e^x(e^x + 1) - (e^x - 1)e^x}{(e^x + 1)^2} = \dfrac{2e^x}{(e^x + 1)^2}$

**69.** $\dfrac{dy}{dx} = \dfrac{2e^{2x}(1 + e^{2x}) - e^{2x} \cdot 2e^{2x}}{(1 + e^{2x})^2} = \dfrac{2e^{2x}}{(1 + e^{2x})^2}$

**70.** $\dfrac{dy}{dx} = e^{-x}(1 - x)$. $e^{-x} > 0$, so $x = 1$ and $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**71.** $\dfrac{dy}{dx} = xe^x(x + 2)$. $e^x > 0$, so $x = 0$ or $x = -2$, giving $(0,\, 0)$ and $\left(-2,\, 4e^{-2}\right)$.

**72.** $\dfrac{dy}{dx} = \dfrac{e^x(x - 1)}{x^2}$, which is 0 when $x = 1$. $y = e$. The point is $(1,\, e)$.

**73.** $a = e^{\ln a}$, so $a^x = e^{x \ln a}$. Since $\ln a$ is a constant, $\dfrac{d}{dx}\left(e^{x \ln a}\right) = \ln a \cdot e^{x \ln a} = (\ln a)\, a^x$.

**74.** The student used the power rule, which only applies when $x$ is the base. Here $x$ is the power, so $\dfrac{d}{dx}\left(2^x\right) = (\ln 2)\, 2^x$.

**75.** $\dfrac{d}{dx}\left(xe^x\right) = e^x + xe^x = (x + 1)e^x$. At $x = 0$: $(0 + 1)e^0 = 1$.

**76.** $\dfrac{dy}{dx} = e^{2x}(1 + 2x)$. At $x = 1$: $e^2(1 + 2) = 3e^2$.

**77.** a) $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$ b) For $x < 1$, $1 - x > 0$ and $e^{-x} > 0$, so $f'(x) > 0$. c) $f'(x) = 0$ when $x = 1$; $f(1) = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

**78.** $4e^{4x-1}$

**79.** $3\left(e^x + 2\right)^2 \times e^x = 3e^x\left(e^x + 2\right)^2$

**80.** $f'(x) = 2xe^{x^2-1}$, so $f'(1) = 2e^0 = 2$.

**81.** $y = e^{-2x}$, so $\dfrac{dy}{dx} = -2e^{-2x}$.

**82.** $-2$

**83.** $27 = 3^3$, so $x + 1 = 3$ and $x = 2$.

**84.** $x = \ln 5$

**85.** $3 + 2 = 5$

**86.** $3\ln x + \ln y$

**87.** $3(2x-5)^2 \times 2 = 6(2x-5)^2$

**88.** $y = (3x+1)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(3x+1)^{-\frac{1}{2}} \times 3 = \dfrac{3}{2\sqrt{3x+1}}$

**89.** $\dfrac{dy}{dx} = 2x - 4 = 2$ at $x = 3$, and $y = -3$. So $y + 3 = 2(x-3)$, i.e. $y = 2x - 9$.

**90.** $\dfrac{dy}{dx} = 3x^2 = 3$, so the normal has gradient $-\frac{1}{3}$.

**91.** $(x-3)(x+2) = 0$, so $x = 3$ or $x = -2$.

**92.** $(2x-1)(x+3) = 0$, so $x = \frac{1}{2}$ or $x = -3$.

**93.** $\frac{1}{2}$

**94.** $x = \dfrac{\pi}{4}$ or $x = \dfrac{5\pi}{4}$

**95.** $72^\circ$

**96.** $x > 3$

**97.** $f(2) = 4$, so $g(4) = 7$.

**98.** Even: $f(-x) = x^4 + x^2 = f(x)$.

**99.** $-5 < 2x + 1 < 5$, so $-3 < x < 2$.

**100.** $0.3 + 0.5 = 0.8$

**101.** $\left(\dfrac{-2+4}{2}, \dfrac{5-1}{2}\right) = (1, 2)$

**102.** $\dfrac{(x-3)(x+3)}{x+3} = x - 3$

**103.** a) $e^4$ b) $3e^0 = 3$ c) $e^3$

**104.** a) $-\dfrac{1}{5e}$ b) $-\dfrac{1}{e^4}$ c) $\dfrac{e}{3}$

**105.** Point $(2, e^2)$, $m = e^2$. $y - e^2 = e^2(x-2)$, so $y = e^2x - e^2$.

**106.** Point $(0, 1)$, $m = 5e^0 = 5$. $y - 1 = 5x$, so $y = 5x + 1$.

**107.** Point $(1, 2e^3)$, $m = 6e^3$. $y - 2e^3 = 6e^3(x-1)$, so $y = 6e^3x - 4e^3$.

**108.** Point $(0, 1)$. Tangent gradient 3, normal gradient $-\frac{1}{3}$. $y - 1 = -\frac{1}{3}x$, so $x + 3y - 3 = 0$.

**109.** Point $(1, e)$. Tangent gradient $e$, normal gradient $-\dfrac{1}{e}$. $e(y-e) = -(x-1)$, so $x + ey - e^2 - 1 = 0$.

**110.** Point $(0,\,1)$. Tangent gradient $-2$, normal gradient $\frac{1}{2}$. $y-1=\frac{1}{2}x$, so $x-2y+2=0$.

**111.** $\dfrac{dy}{dx}=e^{2x}(1+2x)$, so $m=1$ at $x=0$. The tangent is $y=x$.

**112.** Point $\left(2,\,2e^{-2}\right)$. $\dfrac{dy}{dx}=e^{-x}(1-x)$, so $m=-e^{-2}$. $y-2e^{-2}=-e^{-2}(x-2)$, so $y=-e^{-2}x+4e^{-2}$.

**113.** Point $(1,\,e)$. $\dfrac{dy}{dx}=xe^{x}(2+x)$, so $m=3e$. $y-e=3e(x-1)$, so $y=3ex-2e$.

**114.** Point $(-1,\,e)$. $\dfrac{dy}{dx}=2xe^{x^2}=-2e$, so the normal gradient is $\dfrac{1}{2e}$. $2e(y-e)=x+1$, so $x-2ey+2e^2+1=0$.

**115.** Point $(1,\,1)$. Tangent gradient $2e^0=2$, normal gradient $-\frac{1}{2}$. $y-1=-\frac{1}{2}(x-1)$, so $x+2y-3=0$.

**116.** Point $(2,\,1)$. Tangent gradient $2(2)e^0=4$, normal gradient $-\frac{1}{4}$. $y-1=-\frac{1}{4}(x-2)$, so $x+4y-6=0$.

**117.** Point $\left(a,\,e^{2a}\right)$; $\dfrac{dy}{dx}=2e^{2x}$, so $m=2e^{2a}$. Tangent: $y-e^{2a}=2e^{2a}(x-a)$. At $y=0$: $-e^{2a}=2e^{2a}(x-a)$. Since $e^{2a}\neq 0$, $x-a=-\frac{1}{2}$, so $x=a-\frac{1}{2}$.

**118.** Point $(a,\,e^{-a})$; $\dfrac{dy}{dx}=-e^{-x}$, so $m=-e^{-a}$. Tangent: $y-e^{-a}=-e^{-a}(x-a)$. At $y=0$: $-e^{-a}=-e^{-a}(x-a)$. Since $e^{-a}\neq 0$, $x-a=1$, so $x=a+1$.

**119.** Tangent: $y-e^{a}=e^{a}(x-a)$. At $x=0$: $y=e^{a}-ae^{a}=(1-a)e^{a}$.

**120.** Tangent: $y-e^{2a}=2e^{2a}(x-a)$. Through $(0,\,0)$: $-e^{2a}=-2ae^{2a}$, and $e^{2a}>0$, so $a=\frac{1}{2}$. The point is $\left(\frac{1}{2},\,e\right)$ with $m=2e$, so the tangent is $y=2ex$.

**121.** $e^{x}=3$, so $x=\ln 3$ and $y=3$. The point is $(\ln 3,\,3)$.

**122.** The tangent gradient is 6: $3e^{3x}=6$, so $e^{3x}=2$ and $x=\frac{1}{3}\ln 2$. Then $y=2$. The point is $\left(\frac{1}{3}\ln 2,\,2\right)$.

**123.** A horizontal tangent has gradient 0. The gradient is $\dfrac{dy}{dx}=e^{x}$, which is greater than 0 for every $x$, so it is never 0.

**124.** For $y=e^{x}$: $m_1=e^{0}=1$. For $y=e^{-x}$: $m_2=-e^{0}=-1$. $m_1m_2=-1$, so the tangents are perpendicular.

**125.** Point $(0,\,1)$; tangent gradient 1, normal gradient $-1$. Normal: $y=-x+1$, which crosses at $(1,\,0)$.

**126.** The $y$-value must come from the original function, $y=e^{2}$, not from the derivative $2e^{2}$. The correct point is $\left(1,\,e^{2}\right)$.

**127.** a) Point $(0,\,1)$, $f'(x)=2e^{2x}$ so $m=2$. Tangent: $y=2x+1$. b) $2x+1=0$, so $x=-\frac{1}{2}$: the point $\left(-\frac{1}{2},\,0\right)$.

**128.** $e^{3x}+3xe^{3x}=e^{3x}(1+3x)$

**129.** $(\ln 4)\,4^{x}$

**130.** $\dfrac{xe^x - e^x}{x^2} = \dfrac{e^x(x-1)}{x^2}$

**131.** $3x^2e^{x^3}$

**132.** $2\left(1-e^x\right) \times \left(-e^x\right) = -2e^x\left(1-e^x\right)$

**133.** $2x = 5$, so $x = \frac{5}{2}$.

**134.** $x = \log_2 10 = \dfrac{\ln 10}{\ln 2}$

**135.** 7

**136.** 4

**137.** $y = \frac{2}{5}x + \frac{1}{5}$, so the gradient is $\frac{2}{5}$.

**138.** Gradient $-\frac{1}{2}$: $y - 3 = -\frac{1}{2}(x+1)$, so $x + 2y - 5 = 0$.

**139.** $\sqrt{3^2+4^2} = 5$

**140.** $4\left(x^2+1\right)^3 \times 2x = 8x\left(x^2+1\right)^3$

**141.** $\dfrac{(x-1)-x}{(x-1)^2} = -\dfrac{1}{(x-1)^2}$

**142.** $x(x-4) = 0$, so $x = 0$ or $x = 4$.

**143.** $x = \dfrac{-2 \pm \sqrt{4+8}}{2} = -1 \pm \sqrt{3}$

**144.** $\dfrac{\sqrt{2}}{2}$

**145.** $\cos x = \frac{1}{2}$, so $x = \dfrac{\pi}{3}$ or $x = \dfrac{5\pi}{3}$.

**146.** $y > 2$

**147.** $y = 3$

**148.** $f^{-1}(x) = \ln x$

**149.** $x - 3 = 5$ or $x - 3 = -5$, so $x = 8$ or $x = -2$.

**150.** $1 - 0.6 = 0.4$

**151.** $x^3 + 6x^2 + 12x + 8$

**152.** $\dfrac{4x^6}{4x^2} = x^4$

**153.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 7$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

**154.** a) $3\ln x + \ln(x+2)$ b) $\ln(x-1) - \ln(x+5)$ c) $2\ln(4x+3)$

**155.** a) Yes (product) b) No (a sum) c) Yes (power)

**156.** a) $\dfrac{2x}{x^2+1}$ b) $\dfrac{3x^2-4}{x^3-4x}$

**157.** $\dfrac{dy}{dx} = \dfrac{4x-5}{2x^2-5x+3}$

**158.** $\dfrac{dy}{dx} = \dfrac{6}{6x} = \dfrac{1}{x}$

**159.** $\dfrac{dy}{dx} = \dfrac{3}{3x+2}$

**160.** $\dfrac{dy}{dx} = \dfrac{-4}{1-4x} = \dfrac{4}{4x-1}$

**161.** $\dfrac{dy}{dx} = 5 \times \dfrac{2}{2x+3} = \dfrac{10}{2x+3}$

**162.** $\dfrac{dy}{dx} = 3x^2 - \dfrac{2}{x}$

**163.** $\dfrac{dy}{dx} = \dfrac{2x}{x^2+1} + 2e^{2x}$

**164.** $y = 3\ln(2x+1)$, so $\dfrac{dy}{dx} = \dfrac{6}{2x+1}$.

**165.** $y = \frac{1}{2}\ln(4x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2} \times \dfrac{4}{4x-1} = \dfrac{2}{4x-1}$.

**166.** $y = 5\ln x$, so $\dfrac{dy}{dx} = \dfrac{5}{x}$.

**167.** $y = \ln x + \ln(x+3)$, so $\dfrac{dy}{dx} = \dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{2x+3}{x(x+3)}$.

**168.** $\dfrac{dy}{dx} = \dfrac{1}{x+2} - \dfrac{1}{x-1} = \dfrac{(x-1)-(x+2)}{(x+2)(x-1)} = \dfrac{-3}{(x+2)(x-1)}$

**169.** $\dfrac{dy}{dx} = \dfrac{3}{3x-1} - \dfrac{1}{x+1} = \dfrac{3(x+1)-(3x-1)}{(3x-1)(x+1)} = \dfrac{4}{(3x-1)(x+1)}$

**170.** $y = 3\ln x + \frac{1}{2}\ln(x+1)$, so $\dfrac{dy}{dx} = \dfrac{3}{x} + \dfrac{1}{2(x+1)} = \dfrac{7x+6}{2x(x+1)}$.

**171.** $y = 2\ln(x+1) - \ln(x-2)$, so $\dfrac{dy}{dx} = \dfrac{2}{x+1} - \dfrac{1}{x-2} = \dfrac{x-5}{(x+1)(x-2)}$.

**172.** $y = \frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(\dfrac{1}{x+1} - \dfrac{1}{x-1}\right) = \frac{1}{2} \times \dfrac{-2}{x^2-1} = -\dfrac{1}{x^2-1}$.

**173.** Differentiate both sides with the chain rule: $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$. Since $e^{\ln x} = x$, this gives $x\dfrac{d}{dx}(\ln x) = 1$, so $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

**174.** $\dfrac{d}{dx}\ln(5x) = \dfrac{5}{5x} = \dfrac{1}{x}$. By the log laws $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant with derivative 0.

**175.** The student forgot to multiply by the derivative of the inside. Correct: $\dfrac{2x}{x^2+1}$.

**176.** $\dfrac{dy}{dx} = \dfrac{2x}{x^2+3} = \dfrac{2}{4} = \dfrac{1}{2}$ at $x = 1$.

**177.** a) $f'(x) = \dfrac{2x-2}{x^2-2x+5}$ b) $x^2 - 2x + 5 = (x-1)^2 + 4 > 0$ for all $x$, so the log is always defined. c) $f'(x) = 0$ when $x = 1$; $f(1) = \ln 4$. The point is $(1,\, \ln 4)$.

**178.** $3e^{3x} - e^{-x}$

**179.** $2xe^x + x^2e^x = xe^x(2+x)$

**180.** $3(\ln 5)\,5^x$

**181.** Point $(0,\, 1)$, $m = 1$: $y = x + 1$.

**182.** Point $(0,\, 1)$, tangent gradient 1, normal gradient $-1$: $y = -x + 1$.

**183.** $\dfrac{dA}{dt} = 10e^{0.1t}$, so 10 at $t = 0$.

**184.** $e^{\ln 9} = 9$

**185.** $x = e^2$

**186.** $x + 1 = 1$, so $x = 0$.

**187.** $\ln 9 + \ln 4 = \ln 36$

**188.** $x = 2^5 = 32$

**189.** $\dfrac{\ln 20}{\ln 5}$

**190.** $5\left(3x^2-1\right)^4 \times 6x = 30x\left(3x^2-1\right)^4$

**191.** $y = x^{\frac{3}{2}}$, so $\dfrac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$.

**192.** $2x - 6 = 4$, so $x = 5$.

**193.** $(x-3)(x-4) = 0$, so $x = 3$ or $x = 4$.

**194.** $\Delta = 1 - 12 = -11 < 0$, so no real roots.

**195.** $\sqrt{3}$

**196.** $x = \dfrac{7\pi}{6}$ or $x = \dfrac{11\pi}{6}$

**197.** $x \leq 4$

**198.** All real $y$.

**199.** $3x > 9$, so $x > 3$.

**200.** $2x - 1 = 5$, so $f^{-1}(5) = 3$.

**201.** $P(A \cap B) = 0.2$, so $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

**202.** $\dfrac{x(x-4)}{(x-4)(x+4)} = \dfrac{x}{x+4}$

**203.** a) $\dfrac{\ln x}{\ln 6}$ b) $\dfrac{\ln\left(x^2-1\right)}{\ln 3}$

**204.** a) $\dfrac{1}{x\ln 4}$ b) $\dfrac{3}{x\ln 5}$ c) $\dfrac{3}{(3x-1)\ln 2}$

**205.** a) Log law b) Chain rule c) Chain rule d) Log law

**206.** $\dfrac{dy}{dx} = 1 \times \ln x + x \times \dfrac{1}{x} = \ln x + 1$

**207.** $\dfrac{dy}{dx} = 5x^4 \ln x + x^4 = x^4(5\ln x + 1)$

**208.** $\dfrac{dy}{dx} = e^x \ln x + \dfrac{e^x}{x} = e^x\left(\ln x + \dfrac{1}{x}\right)$

**209.** $\dfrac{dy}{dx} = 2x\ln(3x) + x^2 \times \dfrac{1}{x} = x\big(2\ln(3x) + 1\big)$

**210.** $\dfrac{dy}{dx} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{\sqrt{x}}{x} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{2}{2\sqrt{x}} = \dfrac{\ln x + 2}{2\sqrt{x}}$

**211.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}\cdot x^2 - \ln x \cdot 2x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$

**212.** $\dfrac{dy}{dx} = \dfrac{1\cdot \ln x - x\cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{\ln x - 1}{(\ln x)^2}$

**213.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}(x+1) - \ln x}{(x+1)^2} = \dfrac{x+1-x\ln x}{x(x+1)^2}$

**214.** $\dfrac{dy}{dx} = 2\ln x \times \dfrac{1}{x} = \dfrac{2\ln x}{x}$

**215.** $\dfrac{dy}{dx} = 4(\ln x)^3 \times \dfrac{1}{x} = \dfrac{4(\ln x)^3}{x}$

**216.** $y = (\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{\ln x}}$.

**217.** Point $(1, 0)$, $m = 1$. The tangent is $y = x - 1$.

**218.** Point $\left(\frac{1}{3}, \ln 1\right) = \left(\frac{1}{3}, 0\right)$. $\dfrac{dy}{dx} = \dfrac{1}{x}$, so $m = 3$. $y = 3\left(x - \frac{1}{3}\right) = 3x - 1$.

**219.** Point $(e, 2)$. $\dfrac{dy}{dx} = \dfrac{2}{x}$, so $m = \dfrac{2}{e}$. $y - 2 = \dfrac{2}{e}(x - e)$, so $y = \dfrac{2x}{e}$.

**220.** Point $(e, 1)$, tangent gradient $\dfrac{1}{e}$, normal gradient $-e$. $y - 1 = -e(x - e)$, so $y = -ex + e^2 + 1$.

**221.** Point $(1, 0)$. $\dfrac{dy}{dx} = x(2\ln x + 1)$, so $m = 1$. The tangent is $y = x - 1$.

**222.** Point $(1, 0)$. $\dfrac{dy}{dx} = \ln x + 1 = 1$, so the normal gradient is $-1$. The normal is $y = -x + 1$.

**223.** By the change of base law, $\log_a x = \dfrac{\ln x}{\ln a} = \dfrac{1}{\ln a} \cdot \ln x$. Since $\dfrac{1}{\ln a}$ is a constant, $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{\ln a} \cdot \dfrac{1}{x} = \dfrac{1}{x \ln a}$.

**224.** $\ln(x^2) = 2\ln x$, with derivative $\dfrac{2}{x}$. $(\ln x)^2$ needs the chain rule: $\dfrac{2\ln x}{x}$. In the first the power is inside the log; in the second it is on the whole log.

**225.** Point $(e, 1)$, $m = \dfrac{1}{e}$. $y - 1 = \dfrac{1}{e}(x - e)$ gives $y = \dfrac{x}{e}$, which passes through $(0, 0)$.

**226.** $\dfrac{dy}{dx} = \ln x + 1 = 0$, so $\ln x = -1$ and $x = e^{-1}$. $y = e^{-1}\ln e^{-1} = -e^{-1}$. The point is $(e^{-1}, -e^{-1})$.

**227.** a) $f'(x) = \dfrac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$ b) $1 - \ln x = 0$, so $x = e$; $f(e) = \dfrac{1}{e}$. The point is $\left(e, \dfrac{1}{e}\right)$.

**228.** $\dfrac{2x}{x^2 + 7}$

**229.** $\dfrac{1}{x+1} - \dfrac{1}{x-1} = \dfrac{-2}{x^2 - 1}$

**230.** $e^{-x} - xe^{-x} = e^{-x}(1 - x)$

**231.** $(\ln 10)\, 10^x$

**232.** $\dfrac{1}{2\sqrt{x}} e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$

**233.** Point $(0, 1)$, $m = 2$: $y = 2x + 1$.

**234.** $\dfrac{d}{dx}\left(e^{2x} - e^{-2x}\right) = 2e^{2x} - \left(-2e^{-2x}\right) = 2e^{2x} + 2e^{-2x} = 2\left(e^{2x} + e^{-2x}\right)$

**235.** $(e^x - 1)(e^x - 2) = 0$, so $e^x = 1$ or $e^x = 2$: $x = 0$ or $x = \ln 2$.

**236.** $2x - 1 = 1$, so $x = 1$.

**237.** $x + x = 2x$

**238.** $\frac{1}{2}$, since $4^{\frac{1}{2}} = 2$.

**239.** $\ln\left(\dfrac{18 \times 5}{2}\right) = \ln 45$

**240.** $\dfrac{4(2x-3)-2(4x+1)}{(2x-3)^2} = \dfrac{-14}{(2x-3)^2}$

**241.** $y = 6x^{-\frac{1}{2}}$, so $\dfrac{dy}{dx} = -3x^{-\frac{3}{2}}$.

**242.** Tangent gradient 2, normal gradient $-\frac{1}{2}$: $y-1 = -\frac{1}{2}(x-1)$, so $x+2y-3=0$.

**243.** $x(x-2)(x+2) = 0$, so $x = 0, 2$ or $-2$.

**244.** $\Delta = k^2 - 36 = 0$, so $k = \pm 6$.

**245.** $-\dfrac{\sqrt{3}}{2}$

**246.** $\sin x = \pm\dfrac{1}{\sqrt{2}}$, so $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$.

**247.** $\dfrac{2\pi}{3}$

**248.** All real $x$ except $x = 2$.

**249.** $x+1 \geq 3$ or $x+1 \leq -3$, so $x \geq 2$ or $x \leq -4$.

**250.** $g(4) = 25$, so $f(25) = 5$.

**251.** $0.3 \times 0.5 = 0.15$

**252.** $(x-2)\left(x^2+2x+4\right)$

**253.** a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 1$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

**254.** a) $\cos x - \sin x$ b) $4\sec^2 x$ c) $2x + 3\sin x$

**255.** $\dfrac{dy}{dx} = (2x+3)\cos\left(x^2+3x\right)$

**256.** $\dfrac{dy}{dx} = 3x^2\cos\left(x^3\right)$

**257.** $\dfrac{dy}{dx} = 5\cos(5x-2)$

**258.** $\dfrac{dy}{dx} = -2x\sin\left(x^2+1\right)$

**259.** $\dfrac{dy}{dx} = -4\sin 4x$

**260.** $\dfrac{dy}{dx} = -(-3)\sin(2-3x) = 3\sin(2-3x)$

**261.** $\dfrac{dy}{dx} = 6\sec^2 6x$

**262.** $\dfrac{dy}{dx} = 2x\sec^2\left(x^2\right)$

**263.** $\dfrac{dy}{dx} = -\sec^2(1-x)$

**264.** $\dfrac{dy}{dx} = 15\cos 3x - 8\sin 4x$

**265.** $\dfrac{dy}{dx} = 2 \times \frac{1}{2}\sec^2\dfrac{x}{2} = \sec^2\dfrac{x}{2}$

**266.** $\dfrac{dy}{dx} = -\pi\sin(\pi x) - \pi\cos(\pi x)$

**267.** $\dfrac{dy}{dx} = -2\sin 2x$. At $x = \dfrac{\pi}{12}$: $-2\sin\dfrac{\pi}{6} = -1$.

**268.** $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. At $x = \dfrac{\pi}{4}$: $\dfrac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = 2$.

**269.** $\dfrac{dy}{dx} = 3\cos 3x$. At $x = \dfrac{\pi}{9}$: $3\cos\dfrac{\pi}{3} = \dfrac{3}{2}$.

**270.** $\dfrac{dy}{dx} = \cos x + \sin x = 0$, so $\tan x = -1$: $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$.

**271.** $\dfrac{dy}{dx} = 1 - 2\sin x = 0$, so $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

**272.** $\dfrac{dy}{dx} = 2\cos 2x = 0$, so $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$: $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

**273.** $\tan x = \dfrac{\sin x}{\cos x}$, so $\dfrac{d}{dx}(\tan x) = \dfrac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x} = \sec^2 x$.

**274.** At $x = 0$ the curve rises most steeply, with gradient 1; at $x = \frac{\pi}{2}$ it reaches a maximum, so the tangent is horizontal; at $x = \pi$ it falls most steeply, with gradient $-1$. These are the values of $\cos x$ at 0, $\frac{\pi}{2}$ and $\pi$.

**275.** The student forgot to multiply by the derivative of the angle, 3. Correct: $-3\sin 3x$.

**276.** $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. Since $0 < \cos^2 x \leq 1$ wherever $\tan x$ is defined, the gradient is always at least 1, so it is never 0.

**277.** a) $f'(x) = \cos x - \sqrt{3}\sin x$ b) $\cos x = \sqrt{3}\sin x$, so $\tan x = \dfrac{1}{\sqrt{3}}$: $x = \dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$.

**278.** $\dfrac{5}{5x-2}$

**279.** $\ln x + 1$

**280.** $\dfrac{1}{x\ln 3}$

**281.** $\dfrac{2\ln x}{x}$

**282.** $-3e^{-3x}$

**283.** $3x^2e^{2x} + 2x^3e^{2x} = x^2e^{2x}(3 + 2x)$

**284.** Point $(e^2, 2)$, $m = \dfrac{1}{e^2}$. $y - 2 = \dfrac{1}{e^2}\left(x - e^2\right)$, so $y = \dfrac{x}{e^2} + 1$.

**285.** $\dfrac{5\pi}{4}$

**286.** $210°$

**287.** $-\dfrac{\sqrt{2}}{2}$

**288.** $-\sqrt{3}$

**289.** $x = \dfrac{2\pi}{3}$ or $x = \dfrac{4\pi}{3}$

**290.** $x = \dfrac{\pi}{3}$ or $x = \dfrac{4\pi}{3}$

**291.** $\dfrac{\sin^2 x}{\sin x} = \sin x$

**292.** Amplitude 3, period $\pi$.

**293.** $\ell = r\theta = 6 \times \dfrac{\pi}{3} = 2\pi$ cm

**294.** $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 16 \times \dfrac{\pi}{2} = 4\pi$ cm$^2$

**295.** $32 = 2^5$, so $x + 1 = 5$ and $x = 4$.

**296.** $\ln(2x) = \ln 10$, so $2x = 10$ and $x = 5$.

**297.** $4\left(x^2 - 3\right)^3 \times 2x = 8x\left(x^2 - 3\right)^3$

**298.** Tangent gradient 3, normal gradient $-\frac{1}{3}$: $y - 1 = -\frac{1}{3}(x - 1)$, so $x + 3y - 4 = 0$.

**299.** $(x + 5)(x - 1) < 0$, so $-5 < x < 1$.

**300.** Odd: $\sin(-x) = -\sin x$.

**301.** $\dfrac{6}{36} = \dfrac{1}{6}$

**302.** $2x^2 + 5x - 3$

**303.** a) $(\cos x)^3$ b) $(\sin 4x)^2$

**304.** $\dfrac{dy}{dx} = \sin x + x\cos x$

**305.** $\dfrac{dy}{dx} = 2x\cos x - x^2\sin x = x(2\cos x - x\sin x)$

**306.** $\dfrac{dy}{dx} = 3x^2 \tan x + x^3 \sec^2 x = x^2\left(3\tan x + x\sec^2 x\right)$

**307.** $\dfrac{dy}{dx} = \dfrac{\sin x}{2\sqrt{x}} + \sqrt{x}\cos x$

**308.** $\dfrac{dy}{dx} = \sin 3x + 3x\cos 3x$

**309.** $\dfrac{dy}{dx} = 2x\cos 2x - 2x^2 \sin 2x = 2x(\cos 2x - x\sin 2x)$

**310.** $\dfrac{dy}{dx} = \cos x \cdot \cos x + \sin x \cdot (-\sin x) = \cos^2 x - \sin^2 x$

**311.** $\dfrac{dy}{dx} = 2\cos x \times (-\sin x) = -2\sin x \cos x$

**312.** $\dfrac{dy}{dx} = 3\sin^2 x \cos x$

**313.** $\dfrac{dy}{dx} = 2\tan x \sec^2 x$

**314.** $\dfrac{dy}{dx} = \dfrac{-x\sin x - \cos x}{x^2}$

**315.** $\dfrac{dy}{dx} = \dfrac{\sin x - x\cos x}{\sin^2 x}$

**316.** $\dfrac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \dfrac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \dfrac{1+\cos x}{(1+\cos x)^2} = \dfrac{1}{1+\cos x}$

**317.** $\dfrac{dy}{dx} = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x)$

**318.** $\dfrac{dy}{dx} = \dfrac{\cos x}{\sin x}$

**319.** $\dfrac{dy}{dx} = -\sin x\, e^{\cos x}$

**320.** Point $(\pi, -\pi)$. $\dfrac{dy}{dx} = \cos x - x\sin x = -1$. $y + \pi = -(x - \pi)$, so $y = -x$.

**321.** Point $\left(\dfrac{\pi}{4}, \dfrac{1}{2}\right)$. $\dfrac{dy}{dx} = 2\sin x \cos x = 2 \times \dfrac{1}{\sqrt{2}} \times \dfrac{1}{\sqrt{2}} = 1$. $y = x - \dfrac{\pi}{4} + \dfrac{1}{2}$.

**322.** Point $(0, 1)$. $\dfrac{dy}{dx} = e^x(\cos x - \sin x) = 1$, so the normal gradient is $-1$: $y = -x + 1$.

**323.** Quotient rule: $u = \cos x$, $u' = -\sin x$; $v = 1 - \sin x$, $v' = -\cos x$.
$\dfrac{-\sin x(1-\sin x) + \cos^2 x}{(1-\sin x)^2} = \dfrac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \dfrac{1-\sin x}{(1-\sin x)^2} = \dfrac{1}{1-\sin x}$

**324.** $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$ (product rule) and $\dfrac{d}{dx}(\cos x) = -\sin x$. Sum: $\sin x + x\cos x - \sin x = x\cos x$.

**325.** Product rule: $u = e^{-x}$, $u' = -e^{-x}$; $v = \sin x + \cos x$, $v' = \cos x - \sin x$.
$-e^{-x}(\sin x + \cos x) + e^{-x}(\cos x - \sin x) = e^{-x}(-2\sin x) = -2e^{-x}\sin x$

**326.** $2\sin x \cos x + 2\cos x(-\sin x) = 0$. This makes sense because $\sin^2 x + \cos^2 x = 1$, a constant.

**327.** $\dfrac{d}{dx}\ln(\sin x) = \dfrac{\cos x}{\sin x} = \cot x$

**328.** The derivative of a product is not the product of the derivatives. By the product rule, $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$.

**329.** $\dfrac{dy}{dx} = 2x\sin x + x^2\cos x$. At $x = \pi$: $0 + \pi^2(-1) = -\pi^2$.

**330.** a) $f'(x) = -e^{-x}\sin x + e^{-x}\cos x = e^{-x}(\cos x - \sin x)$ b) $e^{-x} > 0$, so $\cos x = \sin x$, $\tan x = 1$: $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

**331.** $2x\cos\left(x^2\right)$

**332.** $5\sec^2 5x$

**333.** $-3\sin(3x+1)$

**334.** $-\sin\dfrac{\pi}{6} = -\dfrac{1}{2}$

**335.** $2xe^{x^2}$

**336.** $\dfrac{3x^2}{x^3+1}$

**337.** $3(\ln 2)\,2^{3x}$

**338.** $\dfrac{1}{x\ln 10}$

**339.** $e^x(1+x)$

**340.** $\dfrac{1-\ln x}{x^2}$

**341.** Point $(1,\, e)$, $m = e$: $y - e = e(x-1)$, so $y = ex$.

**342.** Point $(1,\, 0)$, tangent gradient 1, normal gradient $-1$: $y = -x + 1$.

**343.** $-\dfrac{\sqrt{3}}{2}$

**344.** $2x = 0,\ \pi,\ 2\pi$, so $x = 0,\ \dfrac{\pi}{2},\ \pi$.

**345.** $\cos x = -\dfrac{\sqrt{3}}{2}$, so $x = \dfrac{5\pi}{6}$ or $\dfrac{7\pi}{6}$.

**346.** $\sec^2 x$

**347.** $-1 \leq y \leq 3$

**348.** $\dfrac{\pi}{2}$

**349.** $e^x = e^{-1}$, so $x = -1$.

**350.** $2 + \ln x - \ln x = 2$

**351.** $\log_2 16 = 4$

**352.** $5(1-2x)^4 \times (-2) = -10(1-2x)^4$

**353.** $\dfrac{2x(x+1) - x^2}{(x+1)^2} = \dfrac{x^2+2x}{(x+1)^2}$

**354.** $x = \dfrac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$

**355.** The line has gradient $-\frac{3}{4}$, so the perpendicular gradient is $\frac{4}{3}$.

**356.** $(a+1)^2 - 1 = a^2 + 2a$

**357.** $1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$

**358.** $5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$

**359.** a) $(\sin 5x)^{-1}$ b) $\left(\cos\left(x^3\right)\right)^{-1}$

**360.** a) $\dfrac{2}{\sqrt{3}}$ b) $\dfrac{2}{\sqrt{3}}$ c) 1 d) 1

**361.** $\dfrac{d}{dx}(\cos 3x)^{-1} = -(\cos 3x)^{-2} \times (-3\sin 3x) = \dfrac{3\sin 3x}{\cos^2 3x} = 3\tan 3x \sec 3x$

**362.** $y = (\cos x)^{-2}$, so $\dfrac{dy}{dx} = -2(\cos x)^{-3}(-\sin x) = \dfrac{2\sin x}{\cos^3 x} = 2\tan x \sec^2 x.$

**363.** $\dfrac{dy}{dx} = 5\tan x \sec x$

**364.** $\dfrac{dy}{dx} = -\cot x \operatorname{cosec} x + 2x$

**365.** $\dfrac{dy}{dx} = -2\cot x \operatorname{cosec} x$

**366.** $\dfrac{d}{dx}(\sin 2x)^{-1} = -(\sin 2x)^{-2} \times 2\cos 2x = -2 \cdot \dfrac{\cos 2x}{\sin 2x} \cdot \dfrac{1}{\sin 2x} = -2\cot 2x \operatorname{cosec} 2x$

**367.** $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x - 1$

**368.** $\dfrac{dy}{dx} = -4\operatorname{cosec}^2 x$

**369.** $\dfrac{d}{dx}(\tan x)^{-1} = -(\tan x)^{-2}\sec^2 x = -\dfrac{\cos^2 x}{\sin^2 x}\cdot\dfrac{1}{\cos^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$

**370.** $\dfrac{dy}{dx} = \tan x \sec x - \cot x \operatorname{cosec} x$

**371.** $\dfrac{dy}{dx} = -2\operatorname{cosec}^2 x - 3\tan x \sec x$

**372.** $\dfrac{dy}{dx} = \sec^2 x - \operatorname{cosec}^2 x$

**373.** $\dfrac{dy}{dx} = 2x\tan\left(x^2\right)\sec\left(x^2\right)$

**374.** $\dfrac{dy}{dx} = -3\operatorname{cosec}^2 3x$

**375.** $\dfrac{dy}{dx} = -2\cot(2x-1)\operatorname{cosec}(2x-1)$

**376.** $\dfrac{dy}{dx} = \operatorname{cosec} x - x\cot x \operatorname{cosec} x = \operatorname{cosec} x(1 - x\cot x)$

**377.** $\dfrac{dy}{dx} = 2x\cot x - x^2\operatorname{cosec}^2 x = x\left(2\cot x - x\operatorname{cosec}^2 x\right)$

**378.** $\dfrac{dy}{dx} = e^x\sec x + e^x\tan x\sec x = e^x\sec x(1+\tan x)$

**379.** Point $\left(\dfrac{\pi}{3}, 2\right)$. $m = \tan\dfrac{\pi}{3}\sec\dfrac{\pi}{3} = \sqrt{3}\times 2 = 2\sqrt{3}$. $y - 2 = 2\sqrt{3}\left(x - \dfrac{\pi}{3}\right)$, so $y = 2\sqrt{3}x - \dfrac{2\sqrt{3}\pi}{3} + 2$.

**380.** Tangent gradient $-\cot\dfrac{\pi}{4}\operatorname{cosec}\dfrac{\pi}{4} = -\sqrt{2}$, so the normal gradient is $\dfrac{1}{\sqrt{2}}$.

**381.** Point $\left(\dfrac{\pi}{2}, 0\right)$. $m = -\operatorname{cosec}^2\dfrac{\pi}{2} = -1$. The tangent is $y = -x + \dfrac{\pi}{2}$.

**382.** $\sec x = (\cos x)^{-1}$, so $\dfrac{d}{dx}(\sec x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos x} = \tan x\sec x$.

**383.** $\cot x = \dfrac{\cos x}{\sin x}$, so $\dfrac{d}{dx}(\cot x) = \dfrac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$.

**384.** $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x = -\dfrac{1}{\sin^2 x}$. Since $\sin^2 x > 0$ wherever $\cot x$ is defined, the gradient is always negative.

**385.** The derivative of a reciprocal is not the reciprocal of the derivative. Using the chain rule on $(\cos x)^{-1}$ gives $\dfrac{\sin x}{\cos^2 x} = \tan x\sec x$.

**386.** a) $f'(x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos^2 x}$ b) $\sin x = 0$ gives $x = 0$; the point is $(0, 1)$. c) For $0 < x < \dfrac{\pi}{2}$, $\sin x > 0$ and $\cos^2 x > 0$.

**387.** $\sin x + x\cos x$

**388.** $-2\sin x\cos x$

**389.** $3\sec^2 3x$

**390.** $\cos x\, e^{\sin x}$

**391.** $\dfrac{x\cos x - \sin x}{x^2}$

**392.** $\dfrac{2x}{x^2+1}$

**393.** $(\ln 5)\, 5^x$

**394.** $xe^{-x}(2-x)$

**395.** Point $(\pi, 0)$, $m = \cos\pi = -1$: $y = -x + \pi$.

**396.** $\sec^2 \dfrac{\pi}{3} = 2^2 = 4$

**397.** $\tan x = 1$, so $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

**398.** $-\dfrac{\sqrt{3}}{2}$

**399.** $\dfrac{5\pi}{3}$

**400.** 1

**401.** $\operatorname{cosec}^2 x$

**402.** $\sin x = \frac{1}{2}$, so $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

**403.** $x - 2 = 5$, so $x = 7$.

**404.** $e^{2x} = 4$, so $2x = \ln 4$ and $x = \ln 2$.

**405.** $\log_3 1 = 0$

**406.** $12x^3 + \dfrac{2}{x^2}$

**407.** $\dfrac{x}{\sqrt{x^2+4}}$

**408.** $3x^2 - 12 = 0$, so $x = \pm 2$.

**409.** $x < -3$ or $x > 3$

**410.** $1 = 6 + c$, so $c = -5$.

**411.** $x^2 - x - 2 = 0$, so $x = 2$ or $x = -1$: the points $(2, 4)$ and $(-1, 1)$.

**412.** All real $x$ except $x = 0$.

**413.** $\dfrac{3}{36} = \dfrac{1}{12}$

**414.** $27a^6b^3$

**415.** a) $e^x$ b) $\dfrac{1}{x}$ c) $\cos x$ d) $-\operatorname{cosec}^2 x$ e) $(\ln 3)\,3^x$

**416.** a) Product b) Chain c) Quotient d) Term by term

**417.** $\dfrac{dy}{dx} = e^x \cos 2x - 2e^x \sin 2x = e^x(\cos 2x - 2\sin 2x)$

**418.** $\dfrac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$

**419.** $\dfrac{dy}{dx} = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$

**420.** $\dfrac{dy}{dx} = e^{-3x} - 3xe^{-3x} = e^{-3x}(1 - 3x)$

**421.** $\dfrac{dy}{dx} = -e^{-x}\cos x - e^{-x}\sin x = -e^{-x}(\cos x + \sin x)$

**422.** $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}e^x - e^x \ln x}{e^{2x}} = \dfrac{\frac{1}{x} - \ln x}{e^x} = \dfrac{1 - x\ln x}{xe^x}$

**423.** $\dfrac{dy}{dx} = \dfrac{e^{2x}\cos x - 2e^{2x}\sin x}{e^{4x}} = \dfrac{\cos x - 2\sin x}{e^{2x}}$

**424.** $\dfrac{dy}{dx} = \dfrac{2x\ln x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{x(2\ln x - 1)}{(\ln x)^2}$

**425.** $\dfrac{dy}{dx} = -e^x \sin(e^x)$

**426.** $\dfrac{dy}{dx} = \sec^2 x\, e^{\tan x}$

**427.** $\dfrac{dy}{dx} = \dfrac{e^x}{1 + e^x}$

**428.** $\dfrac{dy}{dx} = \dfrac{1 - \sin x}{x + \cos x}$

**429.** $\dfrac{dy}{dx} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

**430.** $\dfrac{dy}{dx} = \dfrac{2 + 2\cos 2x}{2x + \sin 2x}$

**431.** $\dfrac{dy}{dx} = 3\,(e^x + 1)^2 \times e^x = 3e^x\,(e^x + 1)^2$

**432.** $\dfrac{dy}{dx} = 2(\sin x + \cos x)(\cos x - \sin x)$

**433.** $y = (1+\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(1+\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{1+\ln x}}$.

**434.** $\dfrac{\cos x(1+\cos x)^2 + 2\sin^2 x(1+\cos x)}{(1+\cos x)^4} = \dfrac{\cos x + \cos^2 x + 2 - 2\cos^2 x}{(1+\cos x)^3} = \dfrac{(2-\cos x)(1+\cos x)}{(1+\cos x)^3} = \dfrac{2-\cos x}{(1+\cos x)^2}$

**435.** $\dfrac{dy}{dx} = \dfrac{\left(x^2+1\right)^2 - x\cdot 4x\left(x^2+1\right)}{\left(x^2+1\right)^4} = \dfrac{x^2+1-4x^2}{\left(x^2+1\right)^3} = \dfrac{1-3x^2}{\left(x^2+1\right)^3}$

**436.** $\dfrac{dy}{dx} = \dfrac{e^x\left(e^x+1\right)^3 - e^x\cdot 3e^x\left(e^x+1\right)^2}{\left(e^x+1\right)^6} = \dfrac{e^x\left(1-2e^x\right)}{\left(e^x+1\right)^4}$

**437.** $\dfrac{dy}{dx} = e^x(\sin x + \cos x) = 1$ at $x = 0$. The tangent is $y = x$.

**438.** Point $(1, \ln 2)$. $\dfrac{dy}{dx} = \dfrac{2x}{1+x^2} = 1$. $y = x - 1 + \ln 2$.

**439.** $\dfrac{dy}{dx} = e^{2x}(1+2x) = 1$ at $x = 0$, so the normal gradient is $-1$: $y = -x$.

**440.** In $e^x \sin x$ the last operation is multiplying two functions; in $e^{\sin x}$ one function is inside another. $\dfrac{d}{dx}e^x \sin x = e^x(\sin x + \cos x)$ and $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$.

**441.** $\dfrac{d}{dx}\ln(\sec x) = \dfrac{\tan x \sec x}{\sec x} = \tan x$

**442.** The student used the power rule on an exponential. For $e^{f(x)}$ the power stays the same and is multiplied by its derivative: $2xe^{x^2}$.

**443.** $\dfrac{dy}{dx} = \ln x + 1 - 1 = \ln x$, so the gradient at $x = e$ is 1.

**444.** a) $f'(x) = \dfrac{\frac{1}{x}\cdot x^2 - 2x\ln x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$ b) $\ln x = \frac{1}{2}$, so $x = \sqrt{e}$ and $f(\sqrt{e}) = \dfrac{\frac{1}{2}}{e} = \dfrac{1}{2e}$. The point is $\left(\sqrt{e}, \dfrac{1}{2e}\right)$.

**445.** $5e^{5x-1}$

**446.** $y = \ln 3 + 2\ln x$, so $\dfrac{dy}{dx} = \dfrac{2}{x}$.

**447.** $\dfrac{1}{(x+1)\ln 2}$

**448.** $(\ln 7)\, 7^x$

**449.** $2x\sec^2\left(x^2\right)$

**450.** $3\tan 3x \sec 3x$

**451.** $-\cot x \operatorname{cosec} x$

**452.** $-2\operatorname{cosec}^2 2x$

**453.** $2\sin 3x \times 3\cos 3x = 6\sin 3x \cos 3x$

**454.** $\cos x - x\sin x$

**455.** Point $(0,\ 1)$, $m = -1$: $y = -x + 1$.

**456.** Tangent gradient $\cos 0 = 1$, normal gradient $-1$: $y = -x$.

**457.** Quotient rule: $\dfrac{e^x(e^x+1) - (e^x-1)e^x}{(e^x+1)^2} = \dfrac{e^{2x} + e^x - e^{2x} + e^x}{(e^x+1)^2} = \dfrac{2e^x}{(e^x+1)^2}$

**458.** $-\dfrac{\sqrt{2}}{2}$

**459.** $\sqrt{2}$

**460.** $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$

**461.** $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$, so $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

**462.** $x = \ln 3$

**463.** $x = 25$

**464.** $\ln e^2 = 2$

**465.** $-2(3x+2)^{-3} \times 3 = -6(3x+2)^{-3}$

**466.** $\dfrac{2x(1-x) + x^2}{(1-x)^2} = \dfrac{2x - x^2}{(1-x)^2}$

**467.** Point $(-1,\ -2)$, $m = 2(-1) + 3 = 1$: $y = x - 1$.

**468.** $(2x-1)(x-3) = 0$, so $x = \frac{1}{2}$ or $x = 3$.

**469.** $x = 3$, $y = 9 - 18 + 5 = -4$: $(3,\ -4)$.

**470.** $3x - 1 = 5$ or $3x - 1 = -5$, so $x = 2$ or $x = -\frac{4}{3}$.

**471.** $x \geq 3$

**472.** $0.4 + 0.3 - 0.1 = 0.6$

**473.** $\dfrac{-4-4}{3+1} = -2$

**474.** $x^{-6} \times x^7 = x$